CMOS Digital Integrated Circuit Lab

(BECVL0551)



BACHELOR OF TECHNOLOGY

Degree
In

Electronics Engineering (VLSI Design & Technology)

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Submitted to

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NOIDA INSTITUTE OF ENGINEERING & TECHNOLOGY,

GREATER NOIDA

VISION

To be an Institute of academic excellence in the field of education, with future plan of becoming a deemed university, earn name and hence win faith of the society.

MISSION

To impart to its students a high-quality education, develop their skills, broaden their mental horizon and nurture them into competent and talented professionals to meet the challenges of the new millennium.

VISION OF THE DEPARTMENT

To be a renowned Center of Excellence in Electronics and Communication Engineering, developing globally competent ethical resources to serve society.

MISSION OF THE DEPARTMENT

M1: To impart a robust teaching and learning process thriving on qualified, trained resources and state-of-the-art infrastructure.

M2: To promote innovation and research culture by providing students with hands-on experience for solving real-time problems and developing sustainable products and solutions.

M3: To imbibe ethical values, entrepreneurial zeal, and lifelong learning ability to develop future-ready professionals.

PROGRAMME EDUCATIONAL OBJECTIVES

PEO 1: To excel as a professional at work place through continuous learning and ethically perform with entrepreneurial mindset.

PEO 2: To be able to demonstrate high degree of analytical and design ability.

PEO 3: To effectively serve the society by solving the real-time problem and mentoring.

PROGRAMME OUTCOMES

PO1: Engineering Knowledge:

Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions to complex engineering problems.

PO2: Problem Analysis:
Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions:
Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems:
Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions. (WK8)

PO5: Engineering Tool Usage:
Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World:
Analyze and evaluate societal and environmental aspects while solving complex engineering problems for their impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5 and WK7)

PO7: Ethics:
Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws. (WK9)

PO8: Individual and Collaborative Team Work:
Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication:
Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance:
Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects in multidisciplinary environments.

PO11: Life-Long Learning:
Recognize the need for, and have the preparation and ability for independent and life-long learning, adaptability to new and emerging technologies, and critical thinking in the broadest context of technological change. (WK8)

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EXPERIMENT -1

AIM: Introduction of Cadence tool/Micro wind, Schematic Editor, Layout Editor.

TOOLS REQUIRED: PC, CADENCE TOOLS

INTRODUCTION: Cadence Virtuoso is an electronic design automation (EDA) tool used for designing and analysing integrated circuits. It provides different tools for creating circuit schematics, designing layouts, and working with semiconductor devices.

PROCEDURE:

STEP 1: Cadence Setup and Launch

Make sure your system is connected to the network.

To open the Terminal from the desktop, right click on desktop on the file which you like to use.



Figure 1.1: Red Hat desktop

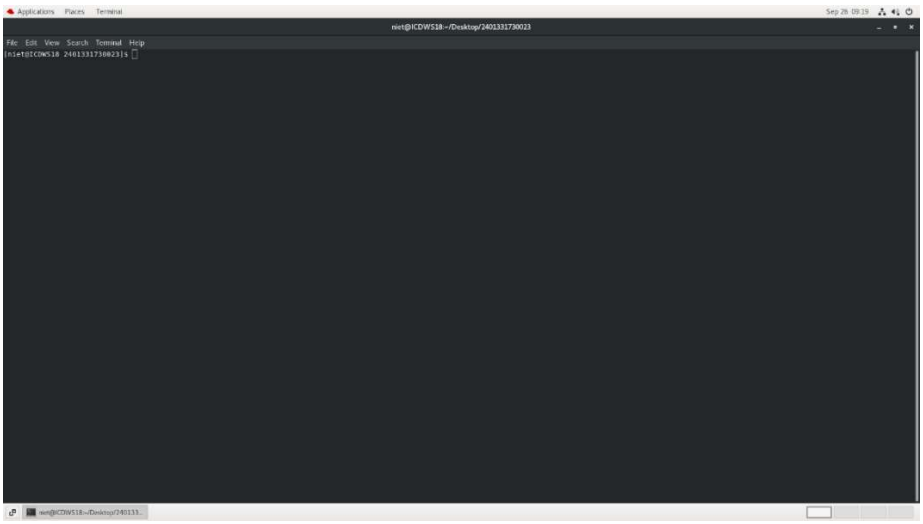


Figure 1.2: Terminal window used to launch

The Cadence environment is launched from the terminal using the commands provided for the laboratory setup.

Commands:

csh ←source /home/install/cshrc ←

A new window will be open then type here

Virtuoso &←

After entering the final command, the Cadence Virtuoso environment opens

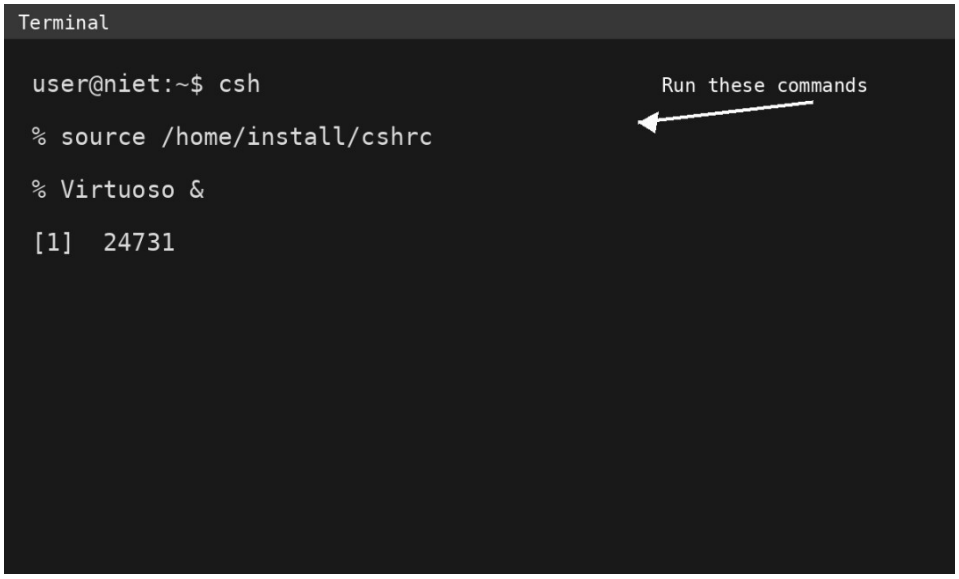


Figure 1.3: Terminal commands used to launch Cadence Virtuoso.



Figure 1.4: Cadence Virtuoso CIW after launch.

STEP 2: Opening Library Manager

After Cadence Virtuoso opens, use the Tools → Library Manager option from the Cadence environment. The Library Manager is used to access libraries, cells and their available views. The basic hierarchy is: File → Library → Cell → View

A library can contain multiple cells, and a cell can contain different views such as schematic

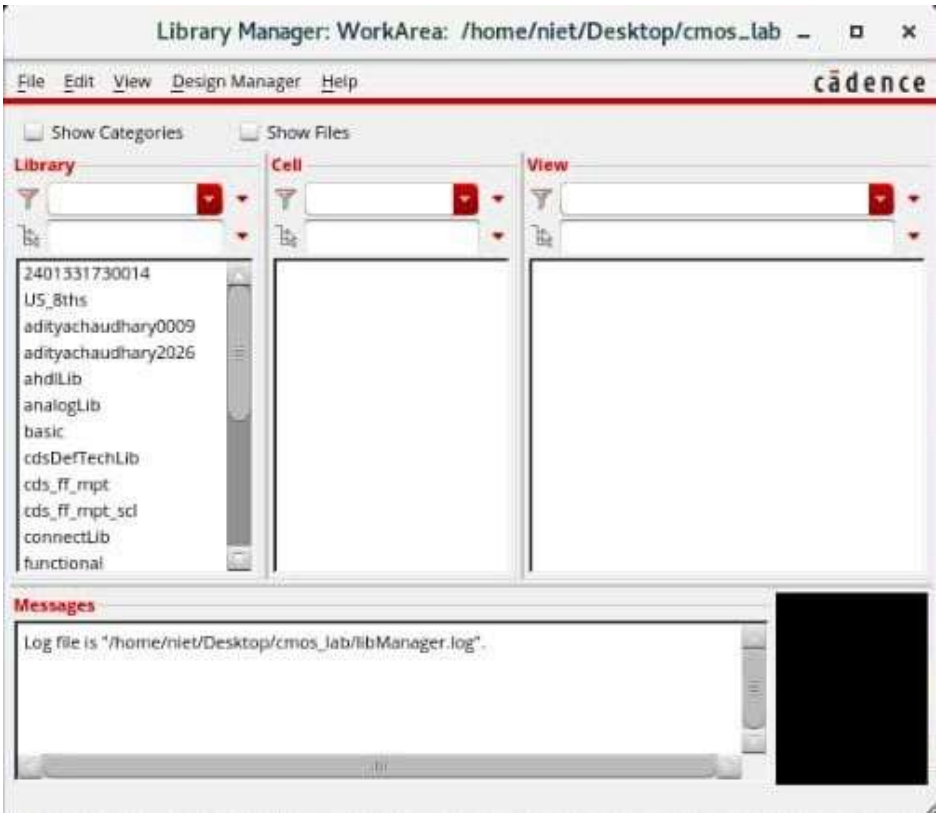


Figure 1.5: Library Manager window.

The hierarchy in Cadence is:
Library (left side) -> Cell (middle) -> View (right).

A library contains multiple cells, and each cell contains multiple views. The libraries that we will use in this tutorial are: -

- **analogLib** - the basic analog components (resistors, capacitors, voltage and current sources, etc.)
- **umc65ll** - the actual components that we will use in the lab (transistors, diodes, op-amps, ...). We won't use them in this tutorial.
- **lab0** - your designs

The views that we will use for each cell are: -

- **schematic** - the actual circuit, the components and interconnections
- **spectre** - the simulation setup
- **symbol** - the appearance of the cell in another schematic view

STEP 3: Create a New Library

In the Library Manager, select **File → New → Library**. Enter the required library name and click OK.

When the Technology File dialog appears, select the technology library configured for the laboratory and confirm the selection. The technology file is collection of information and libraries that define the layers and devices available for a given process technology.



Figure 1.6: New Library dialog.

Step 4: Create a New Schematic Cell View

Select the newly created library in the Library Manager and choose **File → New → Cell View**. Enter the required cell name and select schematic as the view type. Click OK.

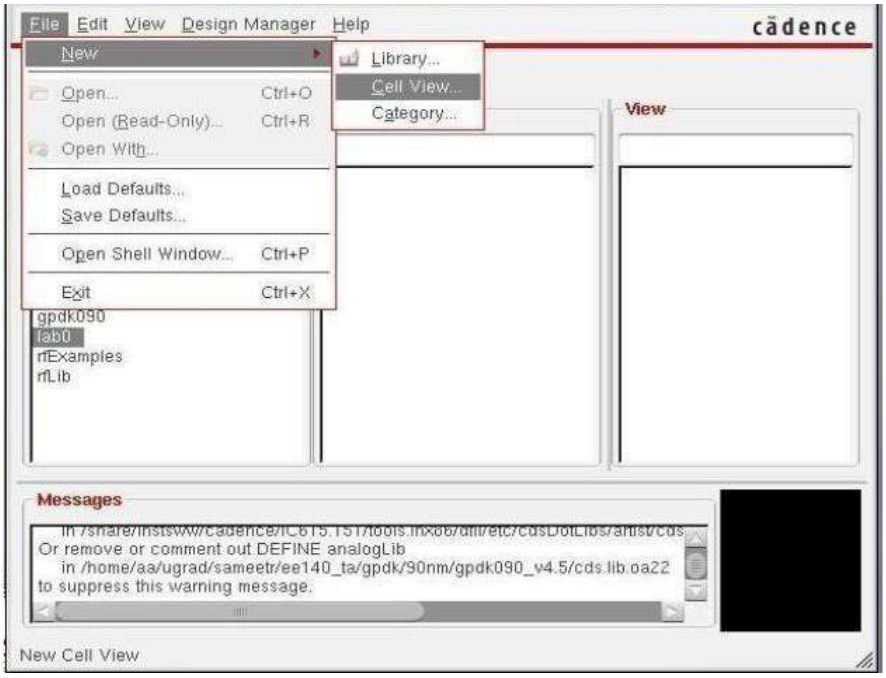


Figure 1.7: File → New → Cell View menu sequence.



Figure 1.8: New File dialog for creating the schematic.

Step 5: Open the Schematic Editor

The Schematic Editor is the workspace used to create a circuit by placing components and connecting them with wires.

Click OK in the new-cell dialog. If the license message appears, select Always when appropriate. The Schematic Editor opens with a grid-based workspace where components and connections are placed.

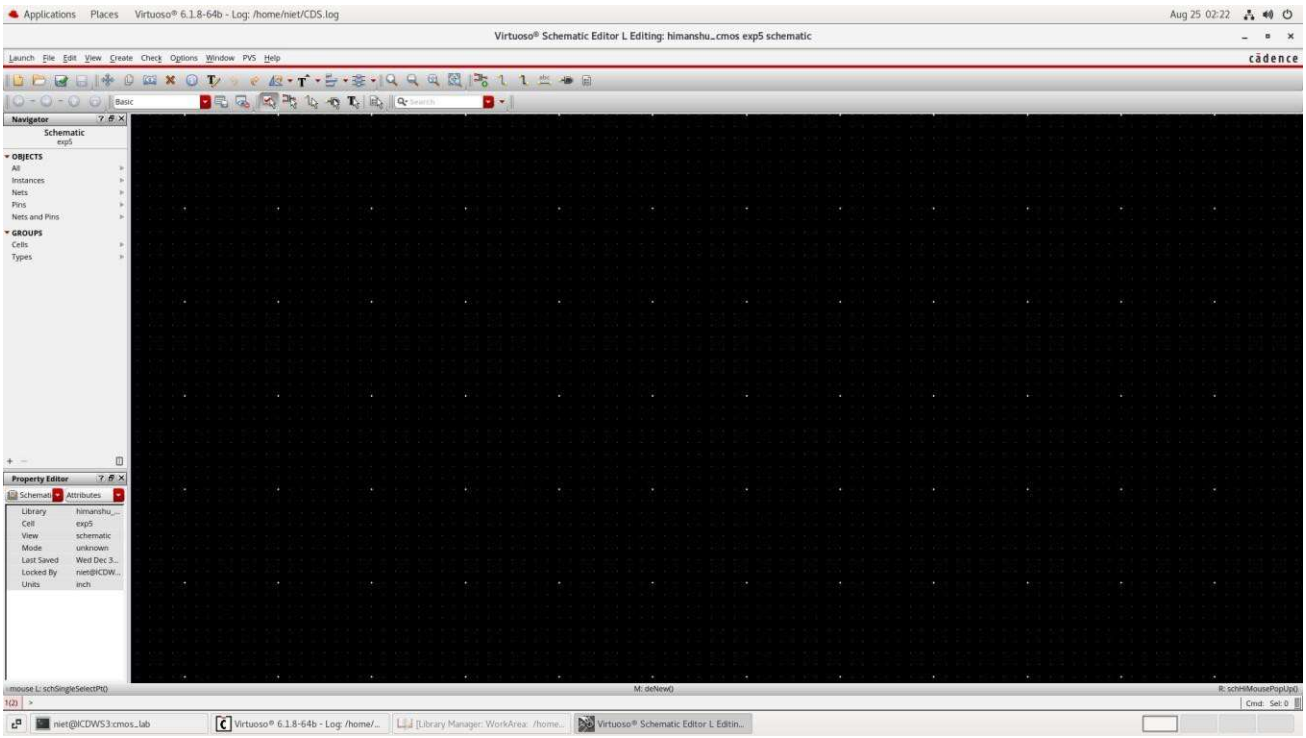


Figure 1.9: Schematic Editor workspace

Step 6: Add Components

To add a component in the Schematic Editor, press

"i". The Add Instance window appears.
Select the required library, cell and symbol view. Components such as gnd, res, cap ,vdc ,vpulse etc. from analogLib may be selected when required.



Figure 1.10: Add Instance window.

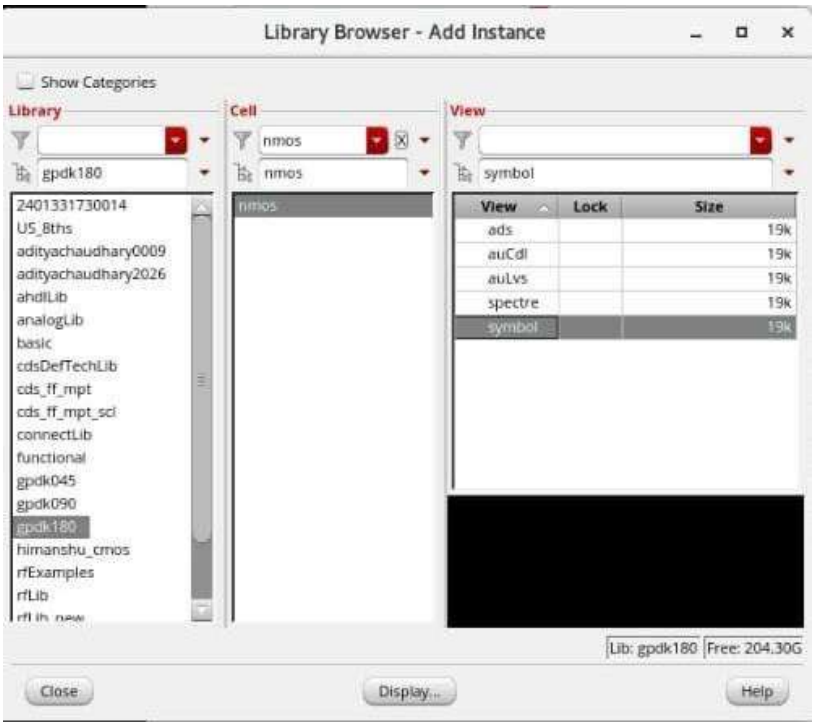


Figure 1.11: Library Browser and component parameter selection.

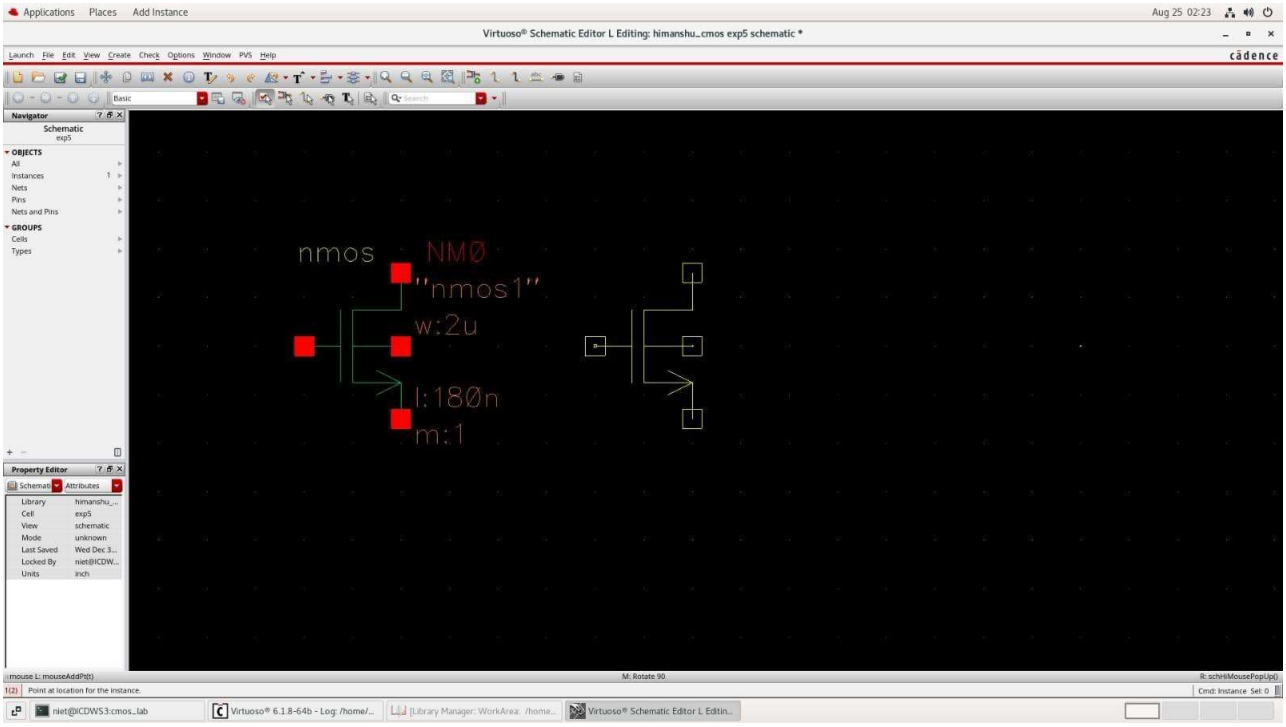


Figure 1.12: Components placed in schematic.

In Cadence you don't have to write the units (Ohms, volts, etc.). For the resistance, type 20k and hit Tab. The Ohms will be automatically completed. The useful prefixes in Cadence are single letters: p - pico, n - nano, u - micro, m - milli, k - kilo, M - mega, G – giga.

. The useful components in the analog library are:

Res	Resistor
Cap	Capacitor
Gnd	Ground
vdc/idc	DC voltage/DC current source
vsin/isin	Sinusoidal voltage/current source
vpulse/ipulse	Square-wave voltage/current source
Iprobe	Current meter

Set the required component properties, place the component in the schematic, and press **Esc** when finish.

Step 7: Make Connections and Check the Schematic

Adding wires and labels: To connect the resistors with a wire, click "w". Click on the first terminal to connect, and then on the next terminal which you want connect.

To create a wire label, click "l" (lowercase L). Type out and click on the wire. Click Esc.

Before continuing, check that the schematic has no unconnected-node errors.

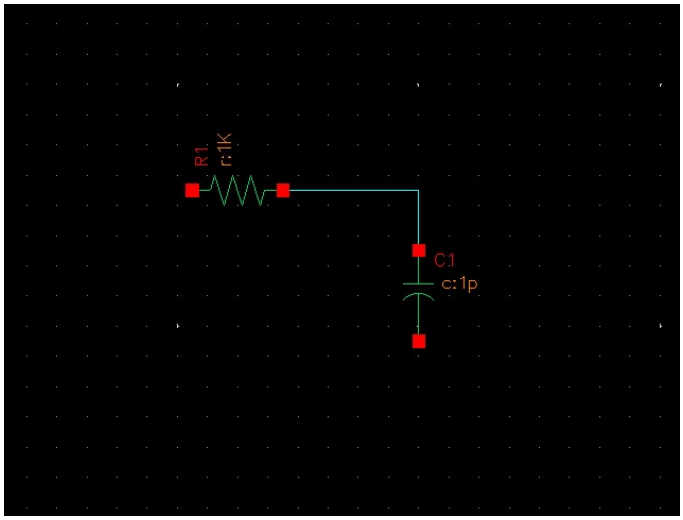


Figure 1.13: Components connected through wire in schematic.

RESULT: The Cadence Virtuoso environment, Library Manager, Schematic Editor and Layout Editor were introduced. The basic procedure for launching Cadence, creating a library and cell view, opening the Schematic Editor, placing components, making connections, checking the schematic, and opening/using the Layout Editor was studied.

EXPERIMENT - 2

Aim: Simulation and analysis of Id-Vgs and Id-Vds characteristics of NMOS.

Software Required : Cadence Virtuoso , Spectre simulator , NMOS transistor model from the configured PDK , DC sweep analysis .

Theory : An NMOS transistor is a voltage-controlled device in which the drain current depends on the gate-to-source voltage (VGS) and drain-to-source voltage (VDS). For a sufficiently large VGS, an inversion channel is formed and drain current increases with the applied drain voltage. With increasing VGS, the conduction capability of the transistor also increases. The DC characteristics are therefore useful for identifying the cut-off, linear and saturation regions.

Circuit Diagram :-

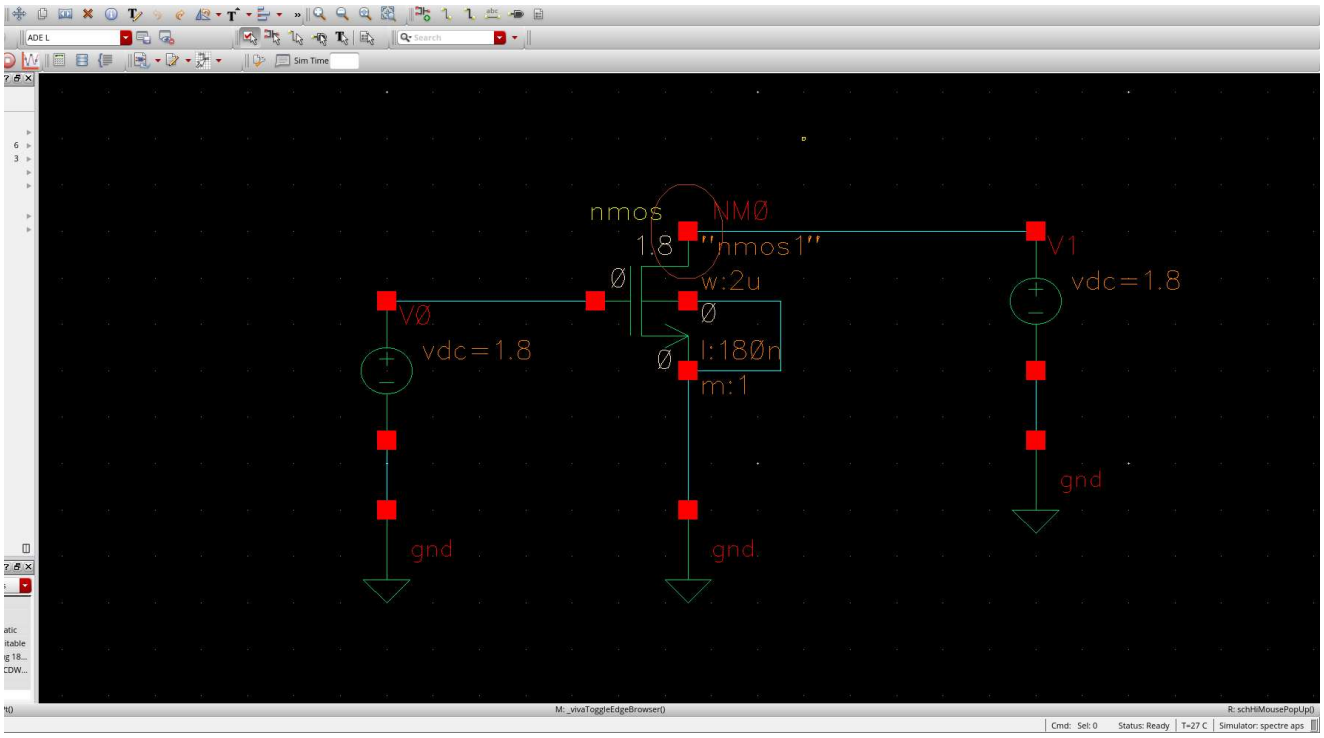


Figure 2.1: NMOS test circuit used for DC characterization.

PROCEDURE:

1. Open the required NMOS schematic in Cadence Virtuoso.
2. Select the wire(narrow) icon in the schematic window and Connect the NMOS source to ground and apply the required DC voltage sources to the gate and drain terminals.
3. Set the transistor dimensions according to the experiment setup and click on check and save.
4. Go to the launch and open ADE L window for showing the graph results.
5. Go to the Analysis →Choose → DC analysis.
6. Then select Component Parameter and define the required sweep range (0 to 1.8 V in the supplied plots).
7. Then click on the select component and select the vdc that is connected to the gate of NMOS and select the dc voltage Component Parameter then click OK.
8. Then go to Output in ADE L window.

Run the simulation and observe the resulting DC characteristic in the Virtuoso Visualization and Analysis window.

Wave form:

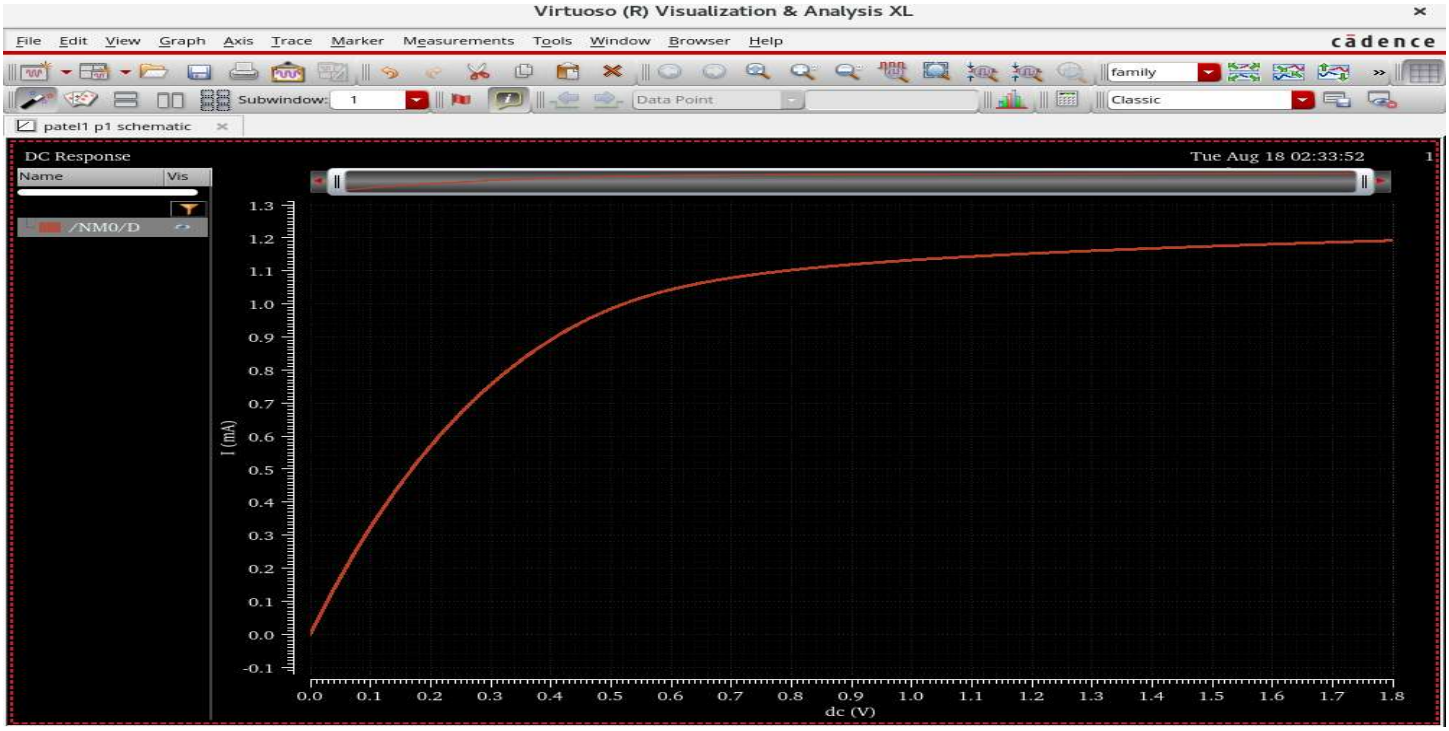
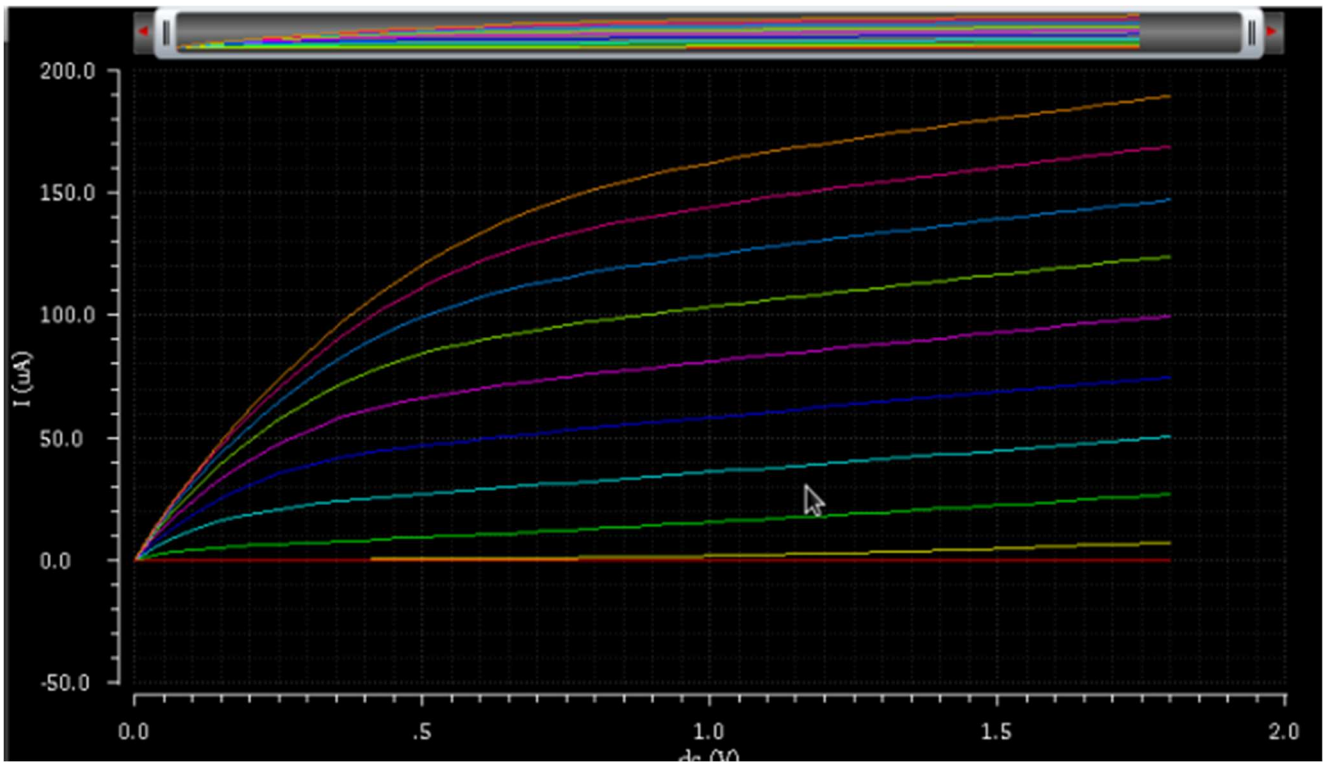


Figure 2.2: NMOS DC characteristic – drain current versus swept DC voltage(Id-Vds).



OBSERVATION:

The simulated NMOS current remains very small at low gate excitation and increases as the device is driven into conduction. The supplied output/transfer plots show the expected nonlinear MOSFET behavior and the increase in current with increasing terminal voltage.

ID–VDS analysis: For output characteristics, keep VGS fixed at selected values and sweep VDS from 0 to 1.8 V. Plot ID versus VDS and identify the linear and saturation regions.

ID–VGS analysis: For transfer characteristics, keep VDS fixed at 1.8 V and sweep VGS from 0 to 1.8 V. Plot ID versus VGS and observe the onset of conduction near the threshold voltage.

Analysis: The simulated curves are used to identify the operating regions of the NMOS and to observe the effect of gate and drain voltages on drain current.

RESULT: The NMOS DC characteristics were obtained successfully using Cadence Virtuoso. The simulation demonstrates the expected increase of drain current with transistor bias and clearly shows the characteristic nonlinear behavior of the NMOS device.

EXPERIMENT – 3

Aim : Simulation and analysis of Id – Vgs and Id - Vds characteristics of PMOS.

Software Required : Cadence Virtuoso , Spectre simulator , NMOS transistor model from the configured PDK , DC sweep analysis .

Theory: A PMOS transistor conducts when the gate voltage is sufficiently lower than the source voltage. Its current-voltage characteristics are complementary to those of an NMOS device. Depending on the chosen current reference direction in the simulator, PMOS drain current may appear as a negative quantity. The DC characteristic therefore reflects both the PMOS conduction behavior and the sign convention used by the simulator.

Circuit Diagram :

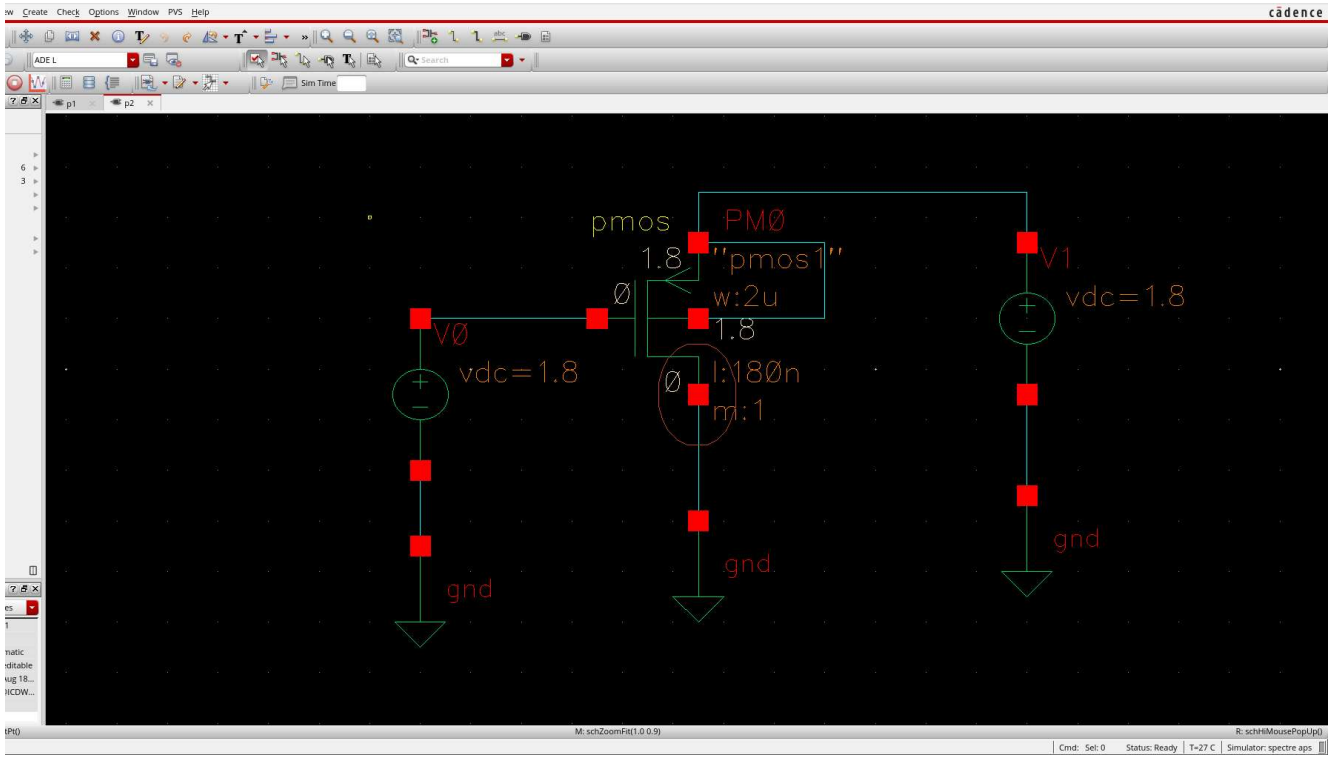


Figure 3.1: PMOS test circuit used for DC characterization.

PROCEDURE:

1. Open the PMOS schematic in Cadence Virtuoso.
2. Select the wire(narrow)and Connect the source and drain terminals according to the PMOS test configuration and provide the required DC bias.
3. Set the PMOS device dimensions as specified in the laboratory setup.
4. Open the Analog Design Environment (Launch →ADE L →Choose)and select DC analysis.
5. Sweep the required DC input variable from 0 to 1.8 V.
6. Then click on the select component and select the vdc that is connected to the gate of PMOS and select the dc voltage Component Parameter then click OK.
7. Select the PMOS drain current node as the output quantity and run the simulation.
8. Observe the current-voltage characteristic in the Virtuoso Visualization and Analysis window.

Wave Form:

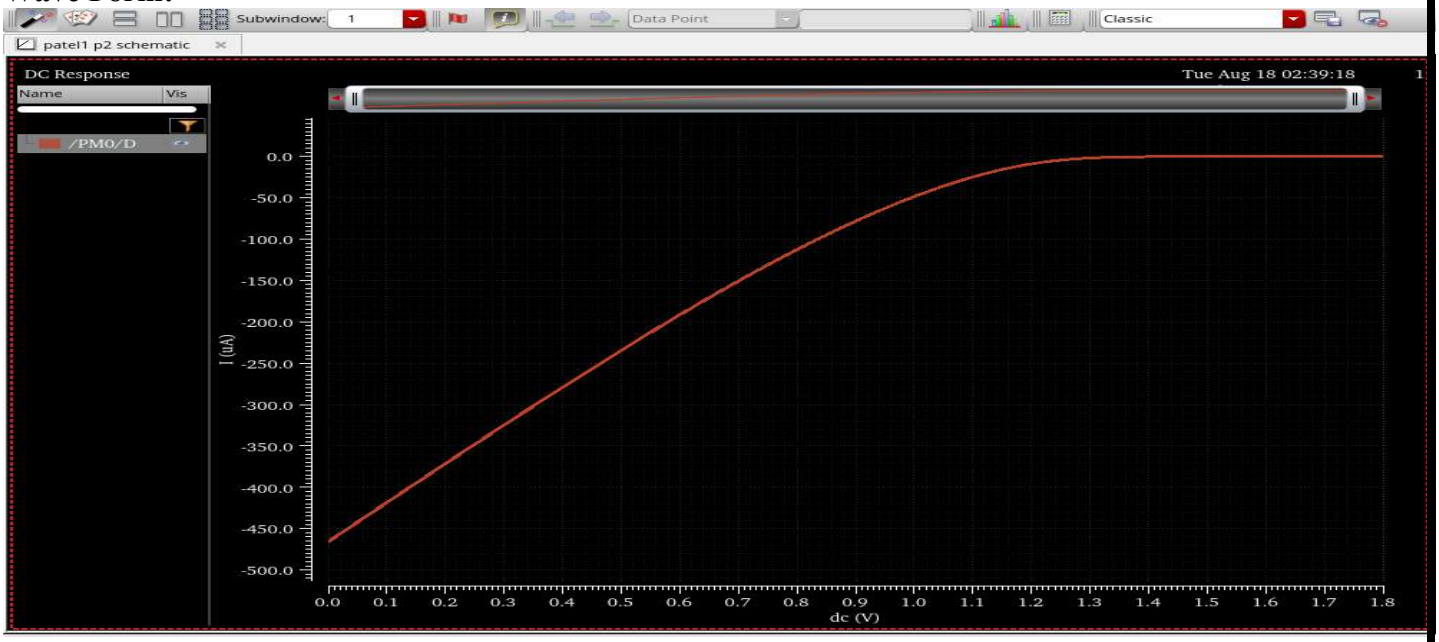


Figure 3.2: PMOS DC characteristic – drain current versus swept DC voltage(Id-Vgs).

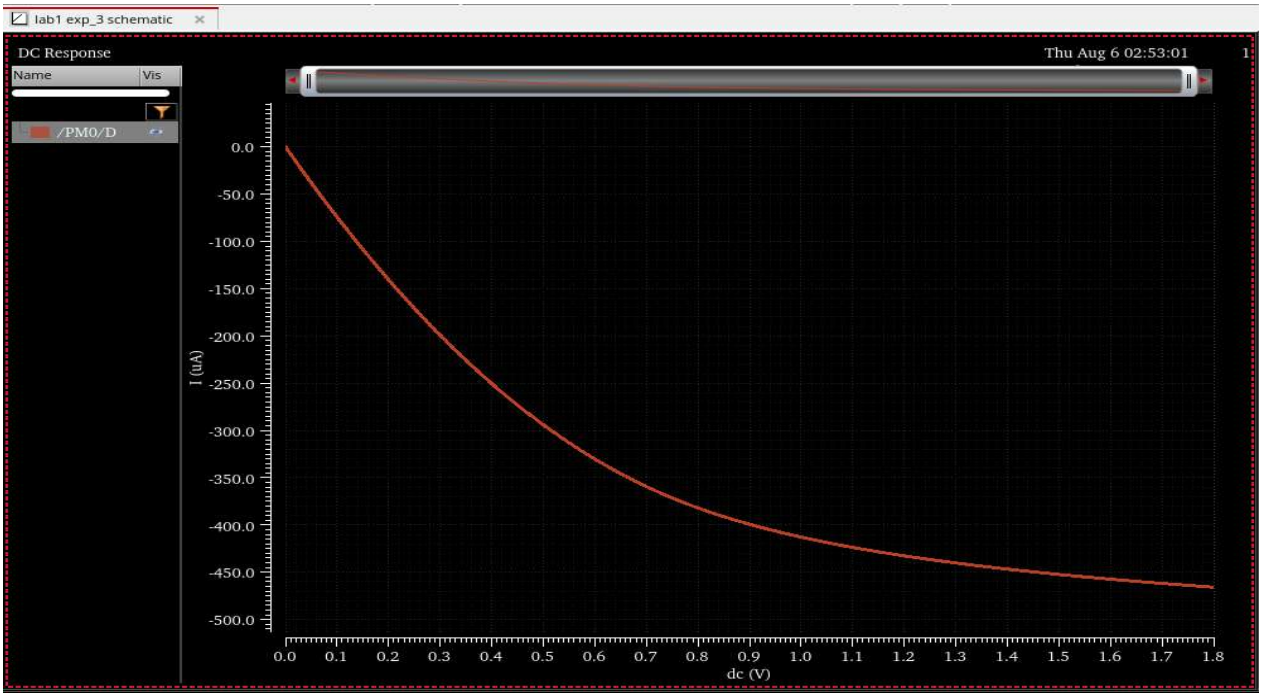


Figure 3.3: PMOS DC characteristic – drain current versus swept DC voltage(Id-Vds).

OBSERVATION: The PMOS simulation produces negative current values over the conducting region because of the current reference direction used in the simulator. The magnitude of current changes significantly as the swept voltage approaches the region where the PMOS device becomes strongly conducting.

RESULT: The PMOS DC characteristics were obtained successfully in Cadence Virtuoso. The simulated response is consistent with PMOS operation and the expected current-sign convention.

EXPERIMENT - 4

Aim : - To analyze the voltage transfer characteristic of resistive load NMOS inverter .

Software Required : Cadence Virtuoso , NMOS transistor , 10 k ohm resistor , DC sweep analysis , 1.8v supply.

Theory : A resistor-load NMOS inverter consists of an NMOS transistor with a resistor connected between the supply voltage and the output node. The input is applied to the NMOS gate. When the input voltage is low, the NMOS is OFF and the resistor pulls the output toward VDD. When the input voltage becomes sufficiently high, the NMOS turns ON and pulls the output toward ground. Thus the circuit provides an inverting voltage transfer characteristic.

Circuit Diagram :-

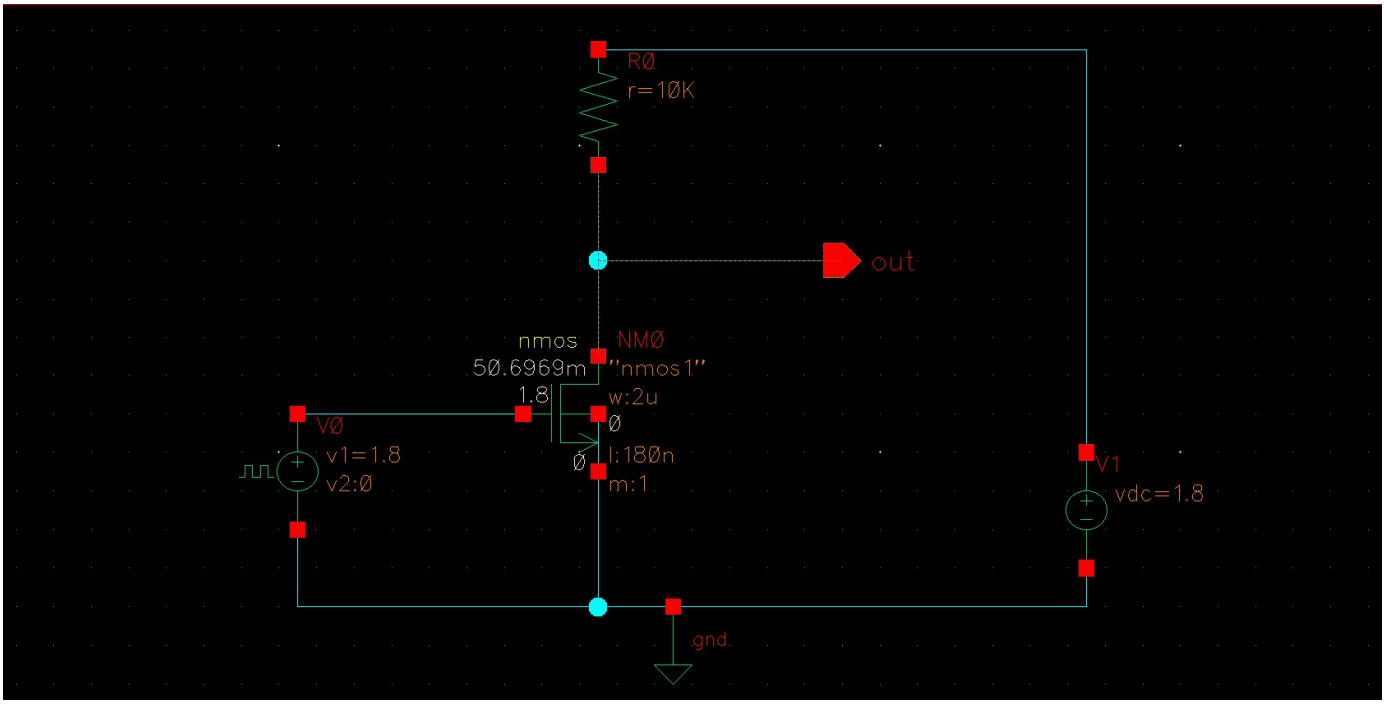


Figure 4.1: NMOS resistor-load inverter schematic (R = 10 kΩ, VDD = 1.8 V).

PROCEDURE:

1. Create/open the NMOS inverter schematic in Cadence Virtuoso.
2. Click the Wire(narrow) icon in the schematic window.
3. Connect a 10 kΩ resistor from the 1.8 V supply to the NMOS drain/output node.
4. Connect the NMOS source to ground and connect the input voltage source to the gate.
5. Label the output node as OUT using PIN (click P) then check and save the schematic.
6. Open the Analog Design Environment(ADE L) and select DC analysis.
7. Sweep the input voltage from 0 to 1.8 V.
8. Then click on the select component and select the vdc that is connected to the gate of NMOS and select the dc voltage Component Parameter then click OK.
9. Go to the Output →To be plotted → Select on schematic →Click on output net ID.
10. Run the simulation and observe the transition region and the high- and low-output voltage levels.

Wave form :

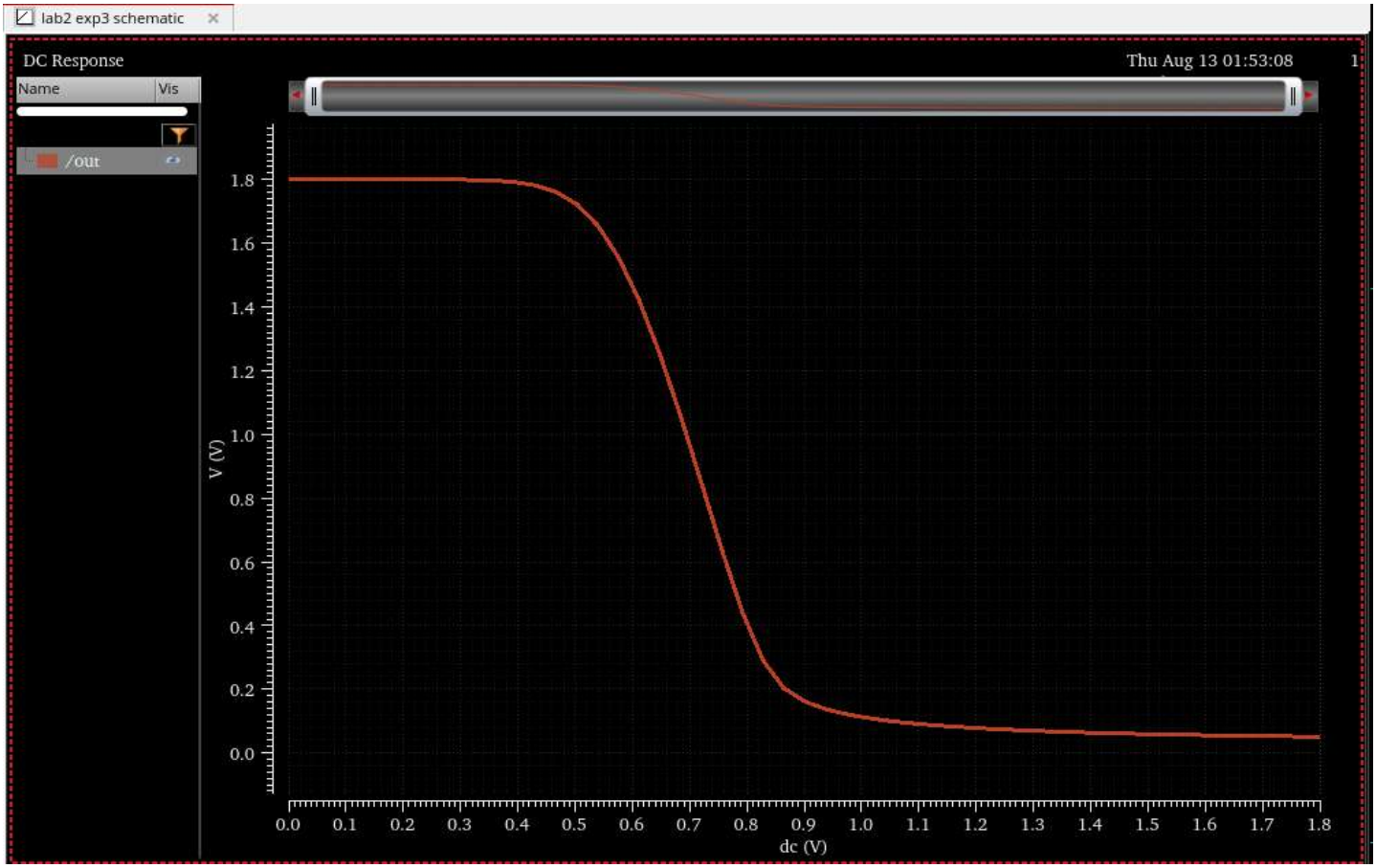


Figure 4.2: Simulated voltage transfer characteristic of resistive-load NMOS inverter (VOUT versus input DC voltage).

OBSERVATION: For a low input voltage, the NMOS is nearly OFF and the output remains close to the supply voltage (approximately 1.8 V). As the input voltage increases, the NMOS conducts more strongly and VOUT falls. At a high input voltage, the output approaches a low-voltage level, confirming the inverting action.

RESULT: The NMOS resistor-load inverter was successfully designed and simulated in Cadence Virtuoso. The obtained VTC confirms the expected inverter operation: low input produces high output, while high input produces low output.

EXPERIMENT - 5

Aim: To analyze the VTC of CMOS inverter.

Software Required: Cadence Virtuoso , NMOS transistor , PMOS transistor , 1.8v supply , V pulse.

Theory: A CMOS inverter consists of a PMOS transistor connected to the supply and an NMOS transistor connected to ground. The gates of both transistors are connected to the input and their drains are connected together at the output. For a low input voltage, the PMOS conducts and the NMOS is OFF, so the output is pulled toward VDD. For a high input voltage, the NMOS conducts and the PMOS is OFF, so the output is pulled toward ground. The VTC is the plot of output voltage (VOUT) versus input voltage (VIN) and shows the transition from logic HIGH to logic LOW.

Circuit Diagram:

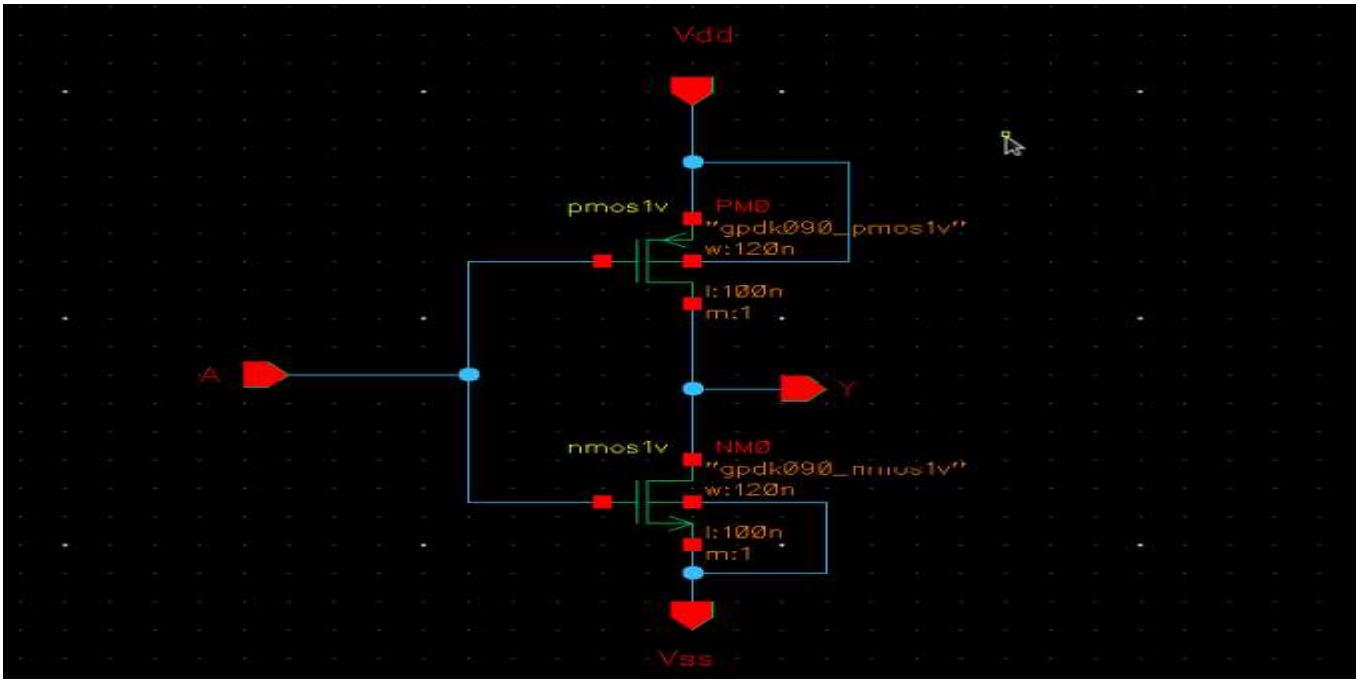


Figure 5.1: CMOS inverter schematic with PMOS and NMOS transistors

PROCEDURE:

1. To create a new library and build a schematic of an inverter.
2. Open the instances and create a CMOS inverter using NMOS and PMOS(**instances →gpdk180 →nmos then pmos**).
3. **Instances →Analog library →vdc.**
4. **Select vdc →click Q(properties) →Dc voltage is 1.8V.**
5. Click the Wire(narrow) icon in the schematic window for connection.
6. Connect the NMOS source to ground and pmos source to the vdd.
7. Label the input(vin) and output node(vout) using PIN (click P) then check and save the schematic.
8. To create the symbol for your inverter go to the inverter schematic window and execute **Create →Cellview →From Cellview**.

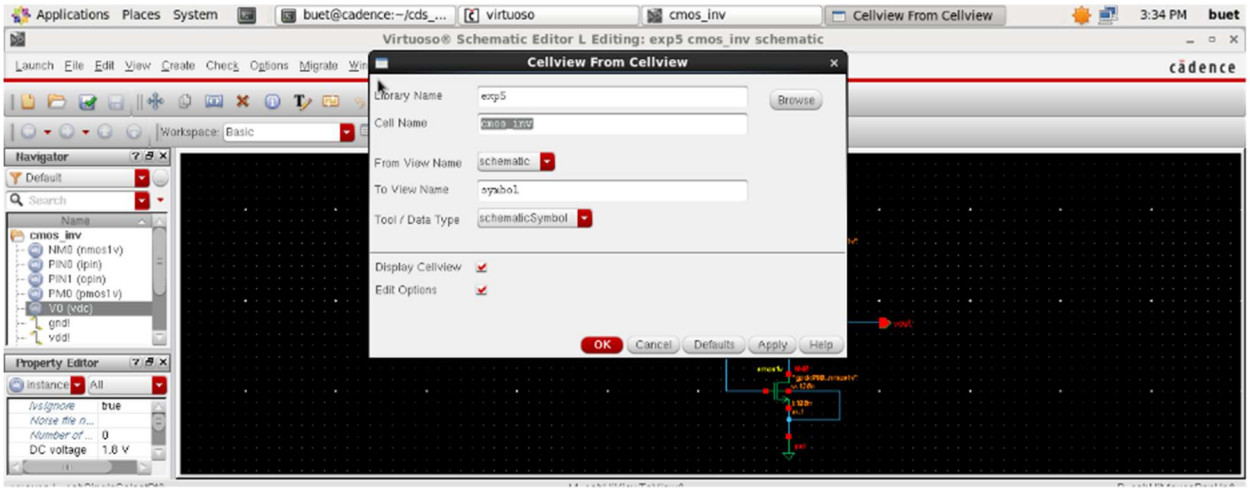


Figure 5.2:Cellview Form to create a symbol of an inverter.

9. Click OK in the Cellview From Cellview form.
10. Then the symbol generation form appears.

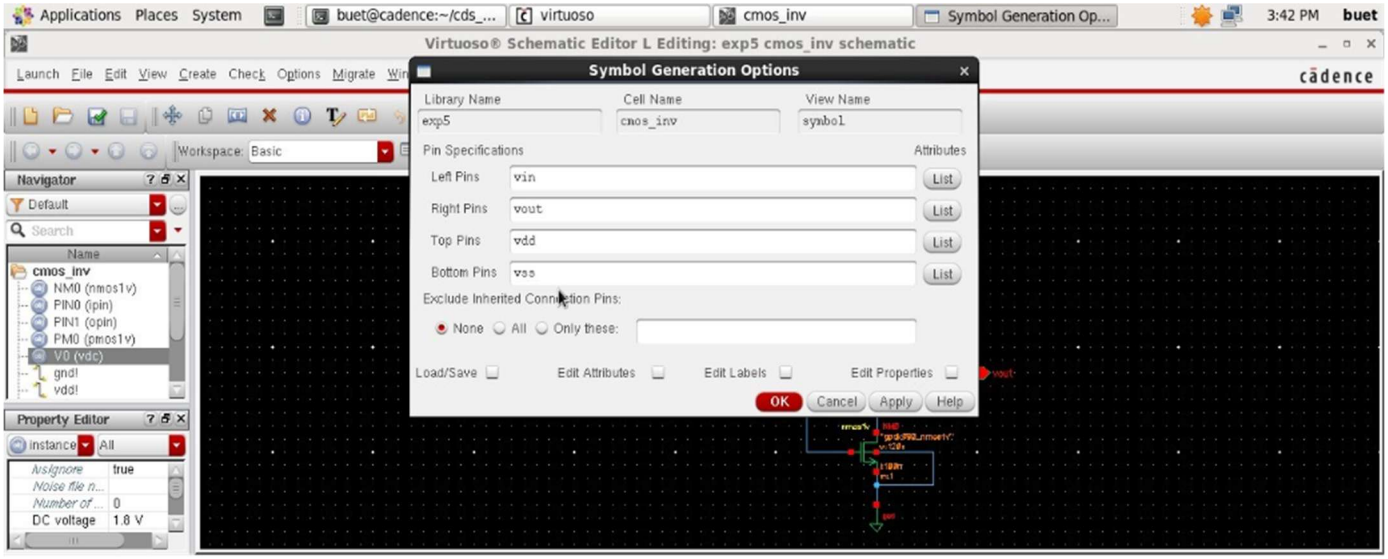


Figure 5.3: Symbol Generation options

11. A new window displays an automatically created inverter symbol .
12. Click left to select the green rectangle →click Delete , similarly select the red rectangle and delete that.
13. Then create a Triangle and a circle at the end of triangle using shapes shown in the virtuoso window.
14. Execute **Create →Selection box.**
15. After creating symbol , click on the save icon to save the symbol.
16. Now building the Inverter_test Design.
17. File →New →cell name(Inverter_test) →OK.
18. A blank schematic window open .
19. Execute **instance →select library name →cell name →symbol →OK.**
20. Open the analog library and add components(vpulse , vdc , gnd).
21. **Select vpulse →click Q →put values(v1=0, v2=1.8V)**.

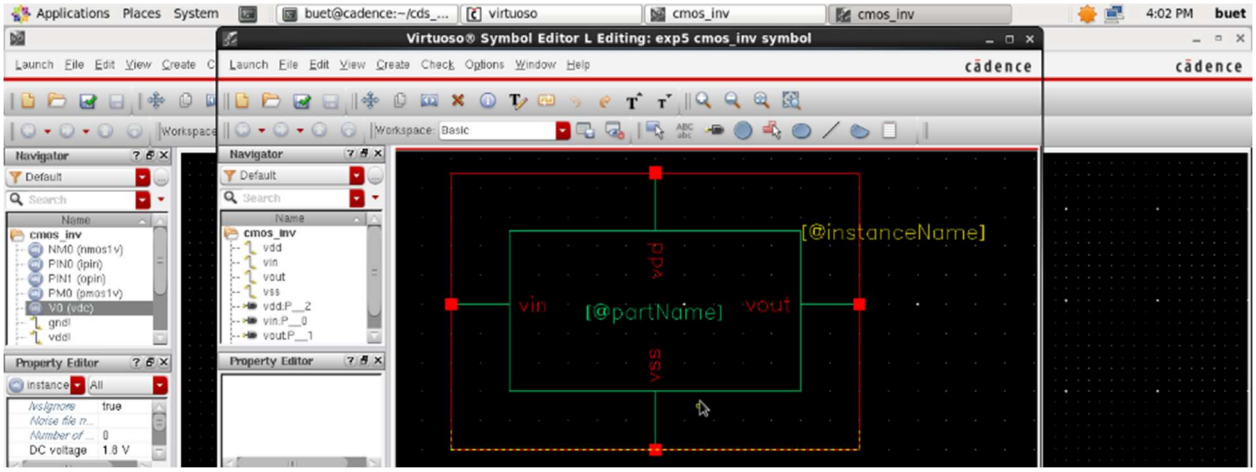


Figure 5.4: Symbol editor window

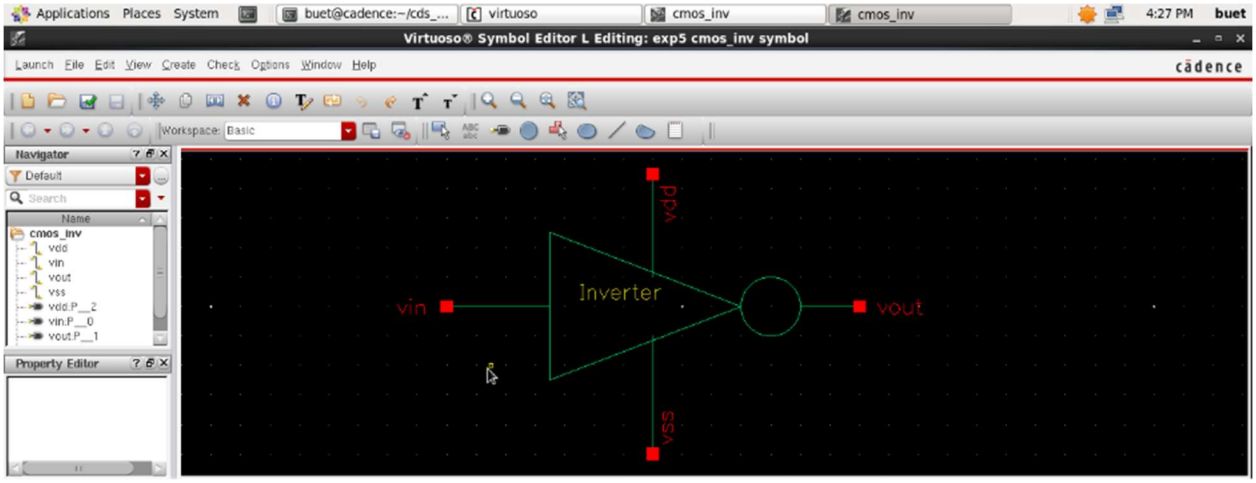


Figure 5.5: Inverter Symbol

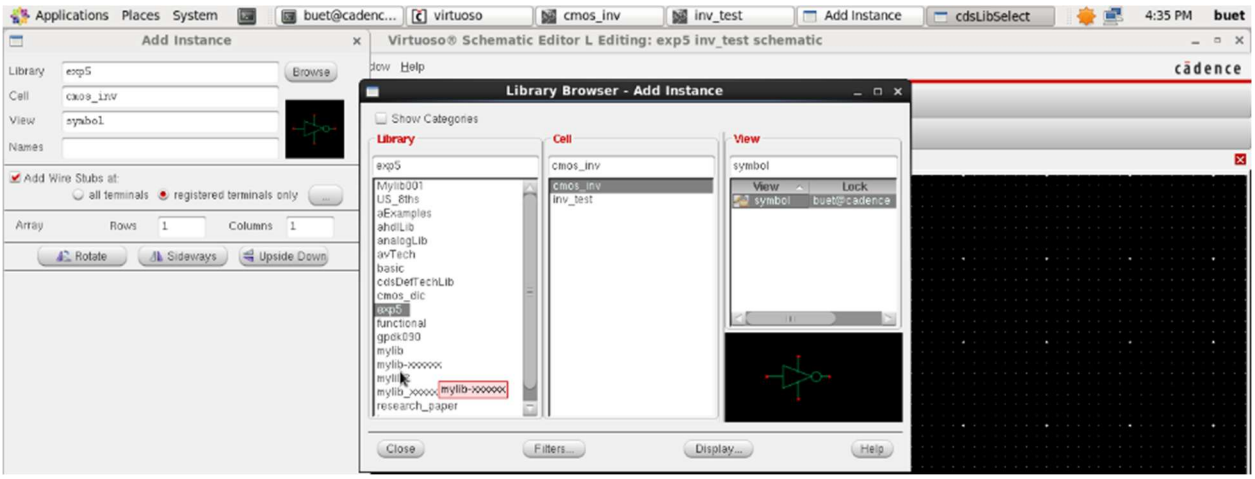


Figure 5.6: Add symbol in inverter test window.

22. Label the input(in) and output(out) node using PIN (click P) then check and save the schematic.
23. Connect the vpulse positive terminal with the vin.

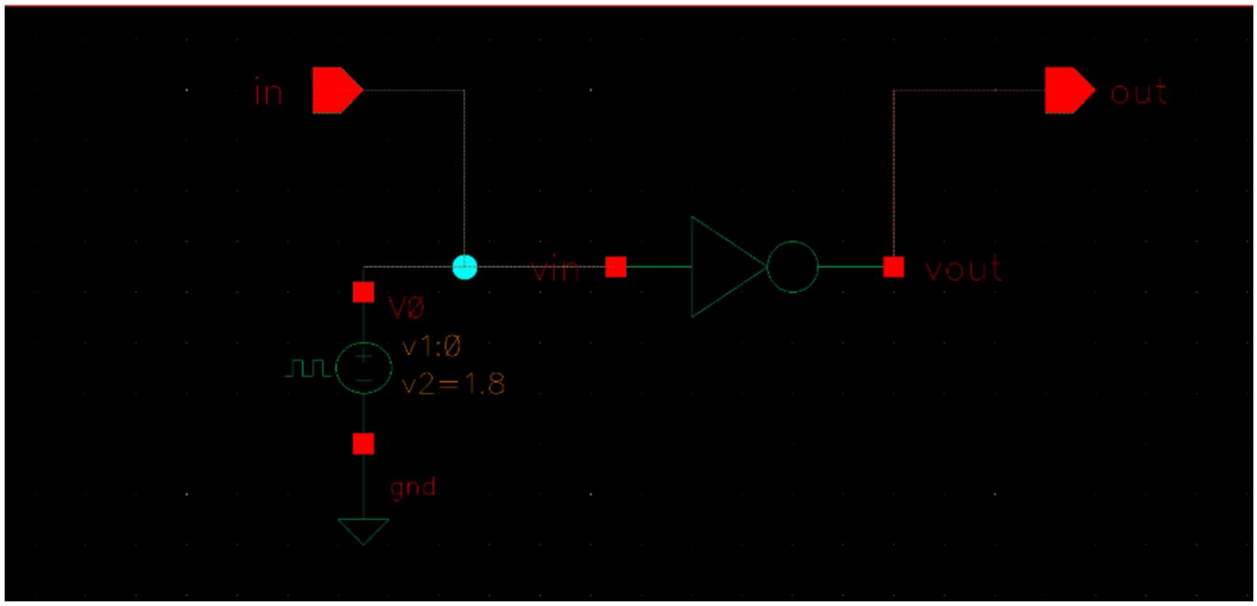
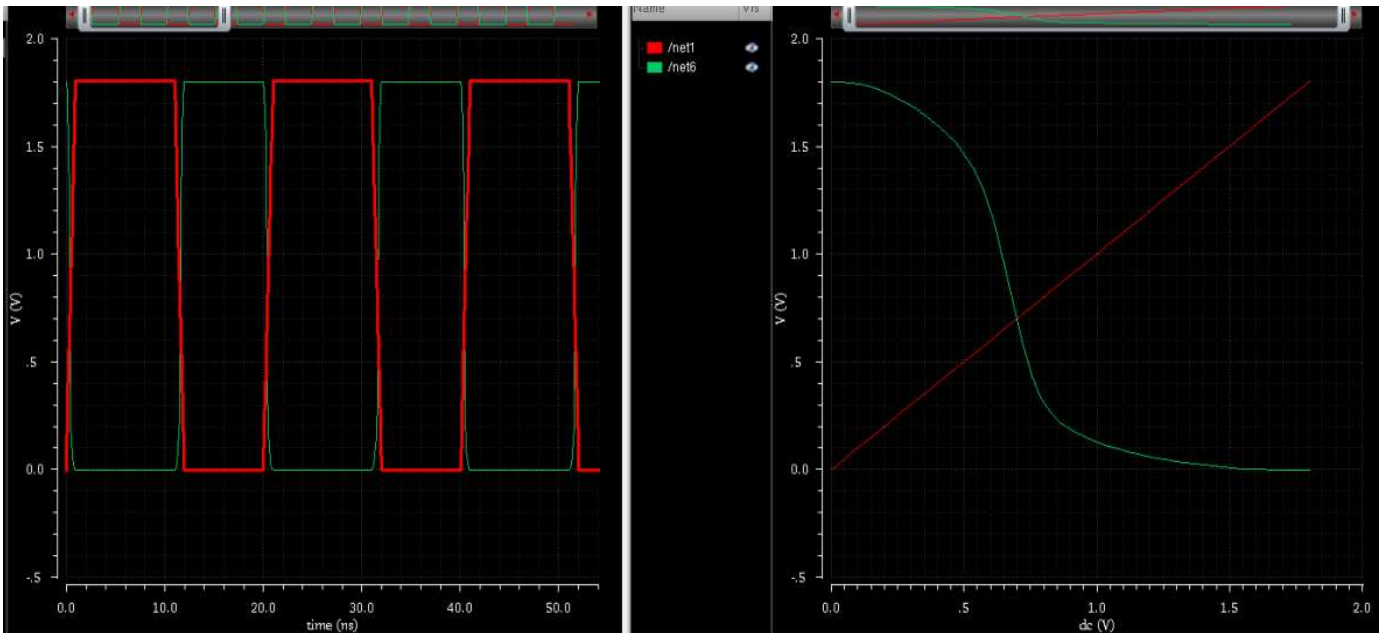


Figure 5.7: Building an Inverter test circuit.

24. Open the Analog Design Environment(ADE L) and select DC analysis.
25. Sweep the input voltage from 0 to 1.8 V.
26. Then click on the select component and select the input pin(in) and select the dc voltage Component Parameter then click OK.
27. Go to the Output →To be plotted → Select on schematic →Click on output pin(out).
28. Execute Simulation – Run the simulation window.



EXPERIMENT – 6

AIM: To design a CMOS inverter circuit in Cadence Virtuoso, analyze its static Voltage Transfer Characteristics (VTC), and extra critical parameters including logic voltage levels (VOH, VOL}, VIH, VIL, switching threshold voltage Vth

SOFTWARE REQUIRED

- **Cadence Virtuoso Schematic Editor** (Circuit schematic creation and design entry)
- **Cadence Spectre Circuit Simulator / ADE L** (DC sweep and operational analysis)
- **Virtuoso Visualization and Analysis Tool** (Waveform plotting and calculator tool)
- **Process Design Kit (PDK):** GPDK 180nm (or GPDK 90nm / 45nm).

THEORY

The CMOS inverter is the fundamental building block of digital integrated circuits. It comprises a PMOS transistor operating as a Pull-Up Network connected to VDD, and an NMOS transistor operating as a Pull-Down Network connect to ground VSS**.**

PROCEDURE

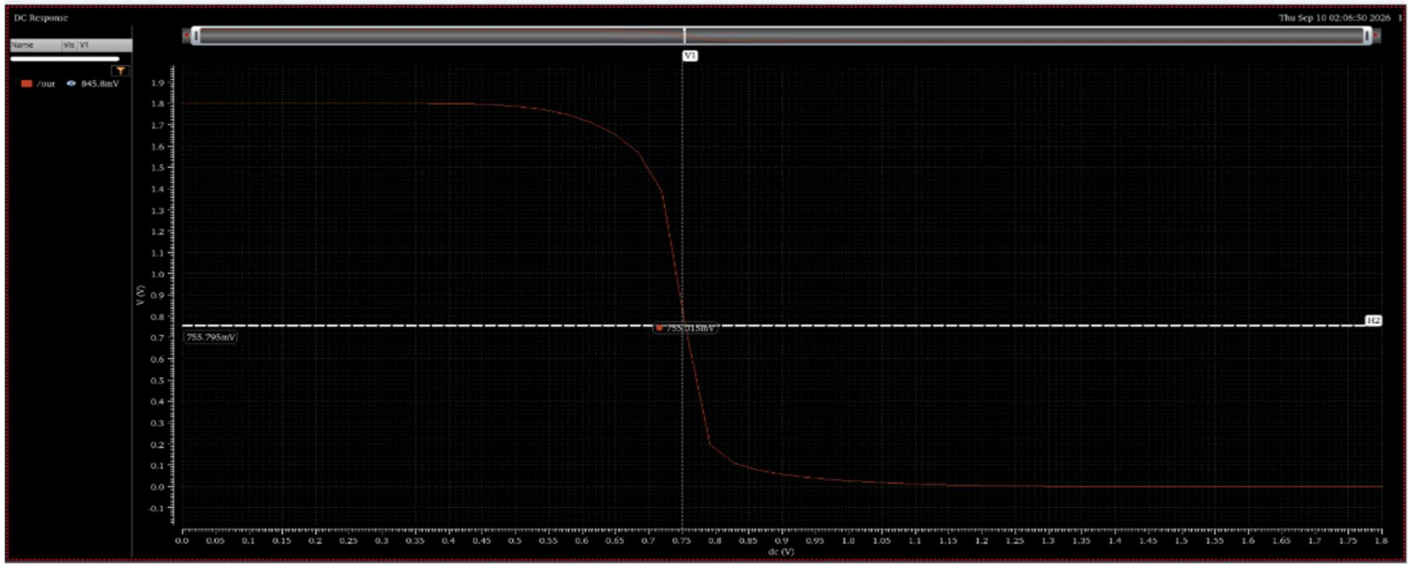
1. **Schematic & Test Bench Setup**:
    a. Open Cadence Virtuoso and launch the Library Manager.
    b. Open the created CMOS Inverter schematic cell view (`gpdk180` process technology) consisting of PMOS (Wp= 2um, Lp= 180nm) and NMOS (Wn= 2um, Ln= 180nm).
    c. Create a test-bench schematic (`Inverter_Test`). Connect VDD = 1.8 V to the PMOS source/substrate, ground to the NMOS source/substrate, and apply a DC source Vin to the common gate terminal.
    d. Perform **Check and Save** (`Shift + X`).

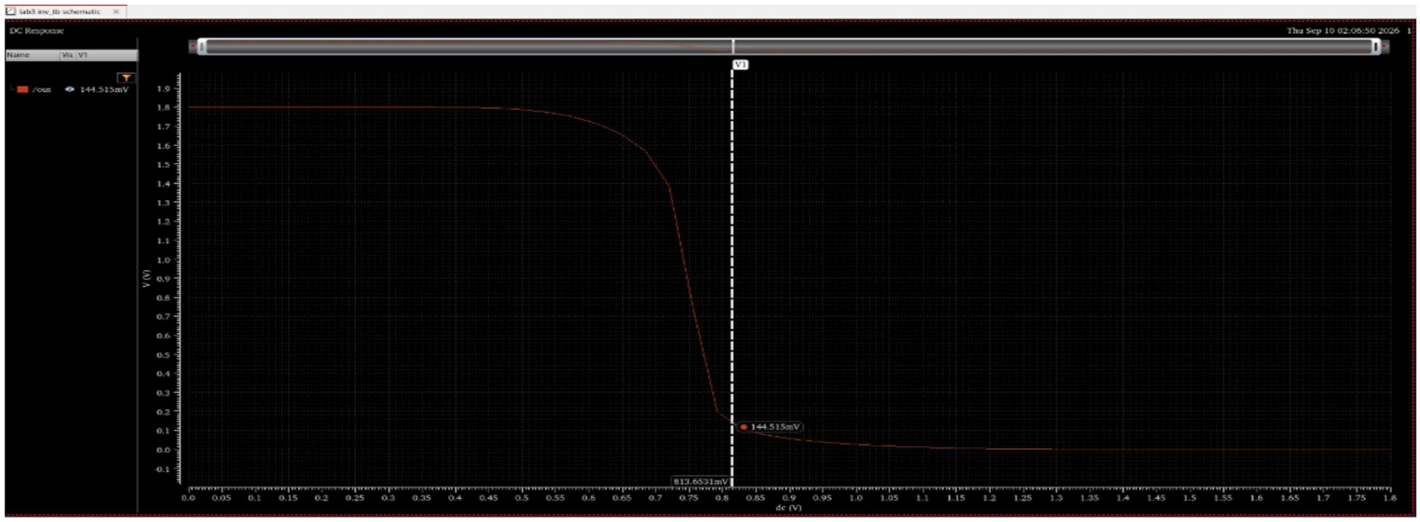
2. **Simulation Setup in Analog Design Environment (ADE L)**:
    a. In the schematic window, select **Launch ADE L**.
    b. Set simulator to **Spectre** and load model library files (`gpdk.scs`).
    c. Select **Analyses Choose DC Analysis**.
    d. Enable **Save DC Operating Point**.
    e. Choose **Component Parameter**, click **Select Component**, pick the V_{in} source on the schematic, and select `dc`.
    f. Set the sweep range: **Start = 0 V**, **Stop = 1.8 V**.

3. **Plotting and Parameter Extraction**:
    a. Go to **Outputs To Be Plotted Select On Schematic** and click the input node (Vin) and output node (Vout).
    b. Execute **Simulatio Netlist and Run**.
    c. Open the **Virtuoso Visualization and Analysis Tool** displaying the VTC curve (Vout vs. Vin).
    d. Use markers (`M`, `A`, `B`) or the **Calculator tool** (`deriv()` function) to extract parameters:
        i. Find VOH and VOL at minimum 0V) and maximum (1.8V) input levels.
        ii. Calculate derivatives {dVout}/{dVin} to pinpoint unity gain points {dVout}/{dVin} = -1) for VIL and VIH
        iii. Find the switching threshold Vth where Vout = Vin.

REGIONS OF OPERATION (CMOS INVERTER VTC)

The CMOS inverter operates across five distinct regions based on the input voltage Vin relative to VDD = 1.8V Vtn (NMOS threshold voltage), and Vtp (PMOS threshold voltage)





Graph 1: Inverter Switching Threshold Voltage (Vth or VM)

- **What it shows:** The operating point where Vin = Vout. At this point, both NMOS and PMOS operate in saturation.
- **Measured Value:** Vertical line V1 and horizontal line H2 intersect on the VTC curve where Vin approx equal V_out = 0.755 V (755.015mV).

Graph 2: Output Low Voltage (VOL)

- **What it shows:** The minimum output voltage produced when the input is fully driven to Logic HIGH (Vin = 1.8 V).
- **Measured Value:** Horizontal line H1 marks the baseline at Vout = 0.0 V.

Graph 3: Input High Voltage (VIH)

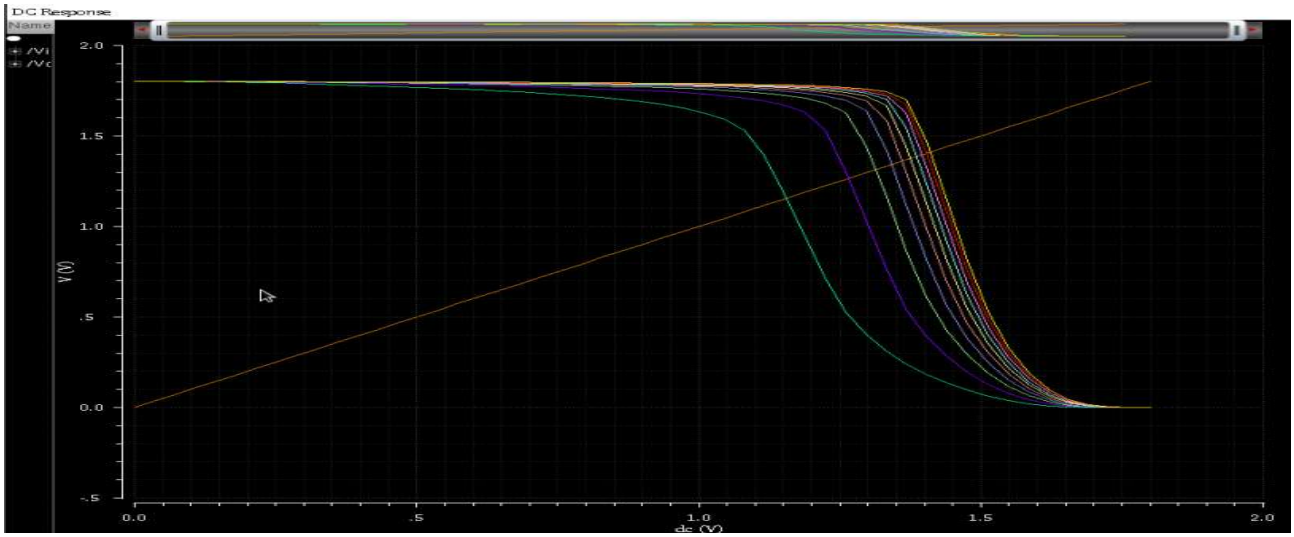
- **What it shows:** The input voltage level where the slope {dVout}/{dVin} = -1 near the end of the transition region. It defines the minimum input voltage recognized reliably as Logic 1.
- **Measured Value:** Vertical line V1 is placed at Vin = 813.65 mV where Vout = 144.51.++

Graph 4: Input Low Voltage (VIL)

- **What it shows:** The input voltage level where the slope{dVout}/{dVin} = -1 near the start of the transition region. It defines the maximum input voltage recognized reliably as Logic 0.
- **Measured Value:** Vertical line V1 is placed at Vin = 679.70mV where Vout = 1.58 V.

Graph 5: Output High Voltage (VOH)

What it shows: The maximum output voltage produced when the input is fully driven to Logic LOW



Measured Value: Horizontal line H1 marks the top output rail at Vout =1.805 V (≈1.8 V).

CALCULATION

- **Output High Voltage (V_{OH})**

$$V_{OH} = V_{DD}$$

Derivation: At $V_{in} = 0$ V, NMOS is OFF ($I_{Dn} = 0$) and PMOS is ON in the triode region. With zero drain current:

$$V_{out} = V_{DD}$$

- **Output Low Voltage (V_{OL})**

$$V_{OL} = 0 \text{ V}$$

Derivation: At $V_{in} = V_{DD}$, PMOS is OFF ($I_{Dp} = 0$) and NMOS is ON in the triode region. With zero drain current:

$$V_{out} = 0 \text{ V}$$

- **Switching Threshold Voltage (V_{th} / V_M)**

At $V_{in} = V_{out} = V_M$, both transistors operate in saturation. Equating drain currents ($I_{Dn} = I_{Dp}$):

$$\frac{1}{2}\beta_n(V_M - V_{Tn})^2 = \frac{1}{2}\beta_p(V_{DD} - V_M - |V_{Tp}|)^2$$

Taking the square root and solving explicitly for V_M:

$$V_{th} = \frac{V_{Tn} + \sqrt{\frac{\beta_p}{\beta_n}}(V_{DD} - |V_{Tp}|)}{1 + \sqrt{\frac{\beta_p}{\beta_n}}}$$

- **Input Low Voltage (V_{IL})**

Occurs at $\frac{dV_{out}}{dV_{in}} = -1$ where NMOS is in saturation and PMOS is in the triode region:

$$I_{Dn} = I_{Dp}$$

$$\frac{\beta_n}{2}(V_{in} - V_{Tn})^2 = \beta_p\left[(V_{in} - V_{DD} - V_{Tp})(V_{out} - V_{DD}) - \frac{1}{2}(V_{out} - V_{DD})^2\right]$$

Differentiating with respect to V_{in} and setting $\frac{dV_{out}}{dV_{in}} = -1$ yields:

$$V_{IL} = \frac{2V_{out} - V_{DD} + V_{Tp} + \left(\frac{\beta_n}{\beta_p}\right)V_{Tn}}{1 + \frac{\beta_n}{\beta_p}}$$

- **Input High Voltage (V_{IH})**
  Occurs at $\frac{dV_{out}}{dV_{in}} = -1$ where NMOS is in the triode region and PMOS is in saturation:

$$I_{Dn} = I_{Dp}$$

$$\beta_n \left[(V_{in} - V_{Tn}) V_{out} - \frac{1}{2} V_{out}^2 \right] = \frac{\beta_p}{2} (V_{in} - V_{DD} - V_{Tp})^2$$

Differentiating with respect to V_{in} and setting $\frac{dV_{out}}{dV_{in}} = -1$ yields:

$$V_{IH} = \frac{V_{DD} - |V_{Tp}| + \left(\frac{\beta_n}{\beta_p}\right)(2V_{out} + V_{Tn})}{1 + \frac{\beta_n}{\beta_p}}$$

MEASURED VALUE:

S. No.	Parameter	Measured Value
1	Output High Voltage (VOH)	1.805 V
2	Output Low Voltage (VOL)	0 V
3	Input Low Voltage (VIL)	679.70 mV
4	Input High Voltage (VIH)	813.65 mV
5	Switching Threshold Voltage (Vth)	755.015 mV

OBSERVATION: The simulated VTC curve shows that the output voltage decreases as the input voltage increases. The measured values are VOH = 1.805 V, VOL = 0 V, VIL = 679.70 mV, VIH = 813.65 mV, and Vth = 755.015 mV.

RESULT: The CMOS inverter was successfully designed and simulated in Cadence Virtuoso. Its static Voltage Transfer Characteristics (VTC) were analyzed, and the parameters VOH, VOL, VIL, VIH, and switching threshold voltage (Vth) were extracted successfully.

EXPERIMENT –7

AIM- Design, simulation, and transient analysis of a 2-input CMOS NOR gate

SOFTWARE REQUIRED- Cadence Virtuoso, Spectre simulator, gpdk090 transistor models (gpdk090_pmos1v, gpdk090_nmos1v), DC Voltage Source (vdc), Pulse Voltage Source (vpulse), and Transient Analysis (tran).

THEORY- A 2-input CMOS NOR gate uses two PMOS transistors in series (pull-up network to Vdd) and two NMOS transistors in parallel (pull-down network to ground). The output V_out is logic HIGH (1V) only when both inputs Va and Vb are LOW (0V), as both PMOS transistors turn ON and both NMOS transistors turn OFF. If either input goes HIGH, the pull-up path turns OFF and at least one NMOS turns ON, pulling Vout to logic LOW (0 V) to implement Vout = {Va + Vb}.

PROCEDURE:

1. Open Cadence Virtuoso and create a new Schematic Cell View for the 2-input CMOS NOR gate.
2. Place two PMOS transistors connected in series between `vdd` and the output node (Vout), forming the pull-up network.
3. Place two NMOS transistors connected in parallel between Vout and `gnd`, forming the pull-down network.
4. Connect input pin Va to the gates of PM0 and NM0, and input pin Vb to the gates of PM1 and NM1.
5. Label input pins (Va, Vb) and output pin (Vout), then Check and Save the schematic.
6. Generate a symbol for the NOR gate using `Create Cellview From Cellview`.
7. Create a testbench schematic (`nor_test`), place the generated NOR gate symbol, and connect a 1.0 V DC supply (`vdc`) to `vdd`.
8. Connect two `vpulse` voltage sources to inputs Va and Vb with staggered time periods to simulate all 4 digital input combinations (00, 01, 10, 11).
9. Open the Analog Design Environment (ADE L), select **Transient Analysis** (`tran`), and set the stop time to {100 ns}.
10. Select nodes Va, Vb, and Vout to be plotted from the schematic, then run the simulation.

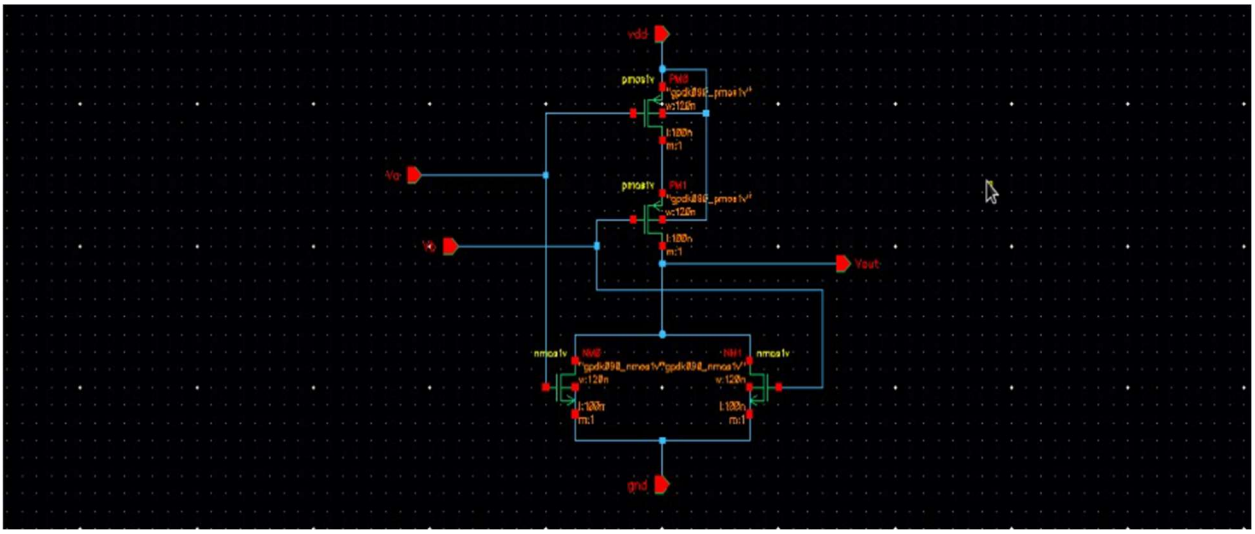


FIG 7.1= schematic of NOR GATE made from cmos inverter

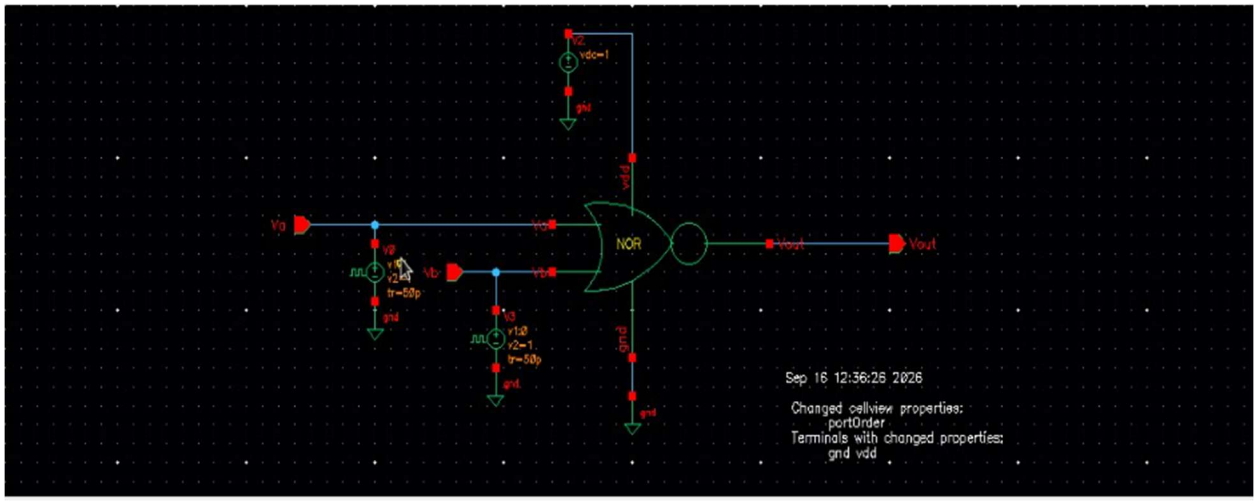
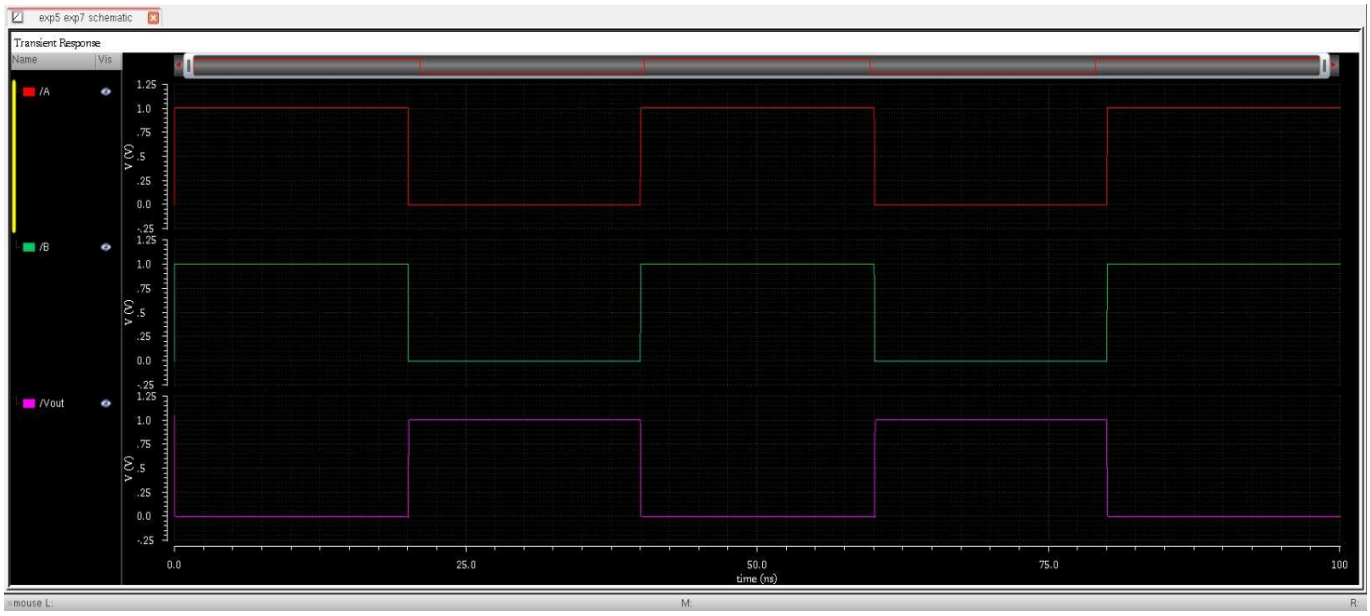


FIG 7.2= pin diagram of nor gate

GRAPH:



Observations from Graph The simulated transient response graph displays the logic behavior over a 100 ns time interval:

- **When Va = 1V (HIGH) and Vb = 1 V (HIGH):** Output Vout = 0V(Logic LOW).
- **When Va = 1V (HIGH) and Vb = 0 V (LOW):** Output Vout = 0t V (Logic LOW).
- **When Va = 0 V(LOW) and Vb = 1 V (HIGH):** Output Vout = 0 V (Logic LOW).
- **When Va = 0 V(LOW) and Vb = 0 V (LOW):** Output Vout approx 1V (Logic HIGH).

The output waveform V_{out} reaches Logic HIGH approx 1.0V only when both inputs Va and Vb are at Logic LOW (0 V) simultaneously, confirming the characteristic behavior and truth table of a 2-input NOR gate.

RESULT: The 2-input CMOS NOR gate was successfully designed and simulated in Cadence Virtuoso. Its transient response was observed, and the output was verified using the truth table.

EXPERIMENT - 8

AIM- To simulate the CMOS 2 input NAND gates and verify the truth tables

TOOLS REQUIRED: PC, CADENCE TOOLS.

PROCEDURE:

- Open **Cadence Virtuoso** and create a new Schematic Cell View for the 2-input CMOS NOR gate.
- Place **two PMOS transistors in series** between VDD and the output node Vout, forming the pull-up network.
- Place **two NMOS transistors in parallel** between Vout and GND, forming the pull-down network.
- Connect input **A** to the gates of the first PMOS and NMOS transistors.
- Connect input **B** to the gates of the second PMOS and NMOS transistors.
- Label the input pins as **A, B** and the output pin as **Vout**. Then perform **Check and Save**.
- Generate the symbol of the NOR gate using **Create → Cellview → From Cellview**.
- Create a testbench schematic and place the generated NOR gate symbol.
- Connect a **1.8 V DC supply (vdc)** to the VDD terminal and connect the ground terminal to GND.
- Connect two **vpulse** voltage sources to inputs **A and B**.
- Open **Analog Design Environment (ADE L)** and select **Transient Analysis (tran)**.
- Select the nodes **A, B and Vout** for plotting.
- Run the simulation and observe the transient waveforms of the two inputs and output.

CIRCUIT DIAGRAM: Two-input NAND Gate Schematic

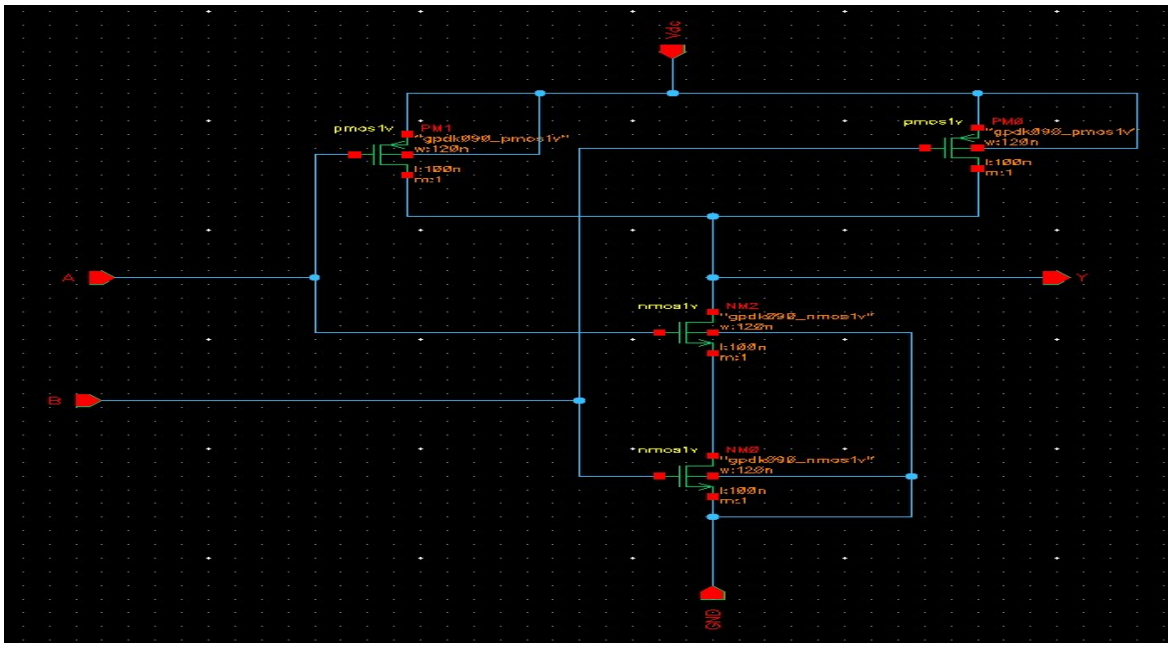


Fig: 8.1 schematic of NAND GATE made from cmos inverter

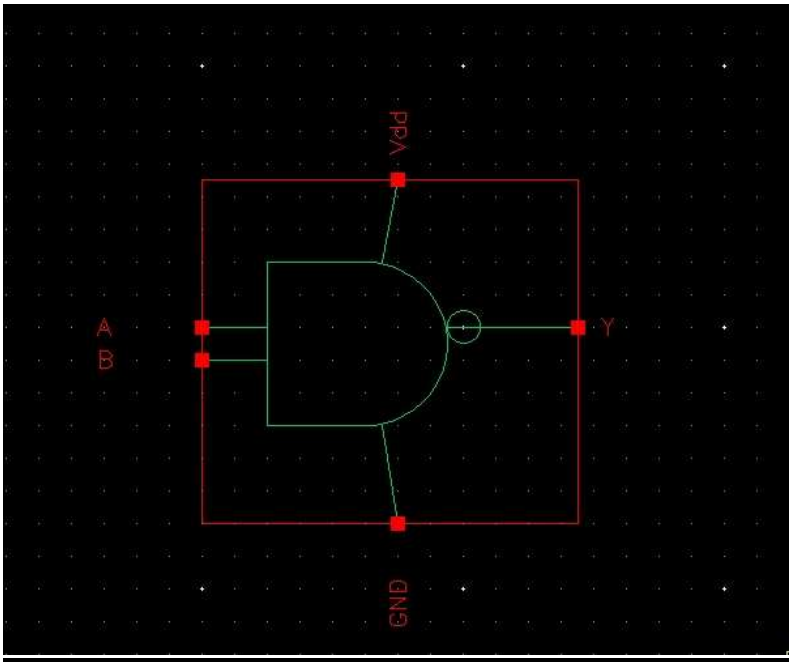


Fig: 8.2 Symbol editor window

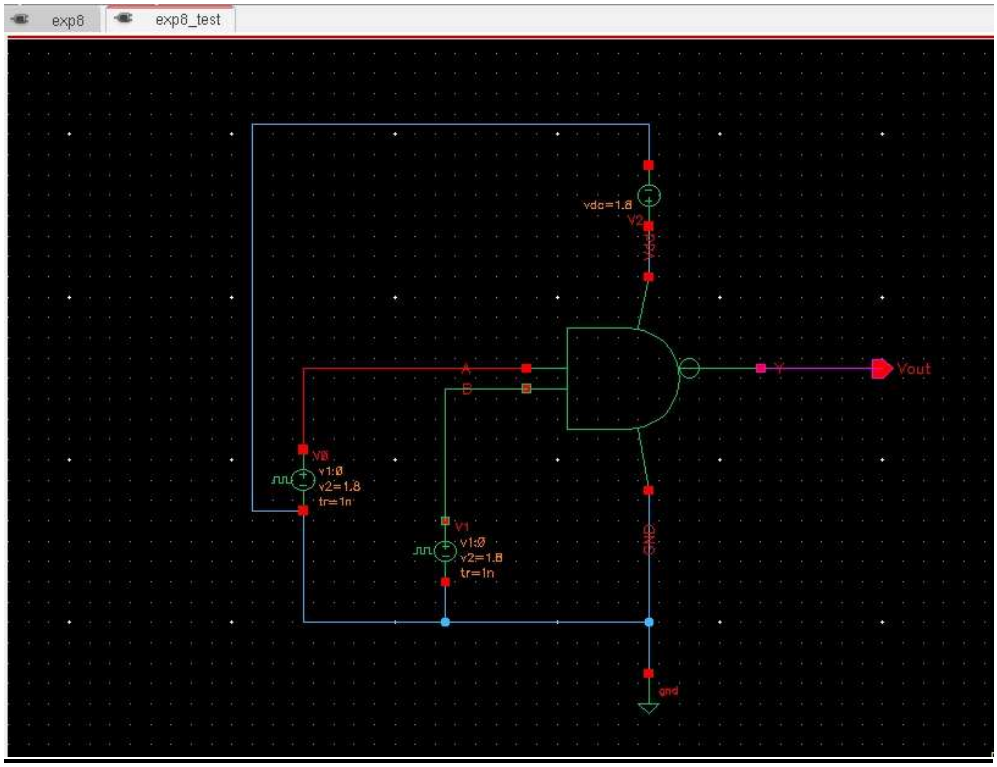


FIG 8.3= pin diagram of nand gate

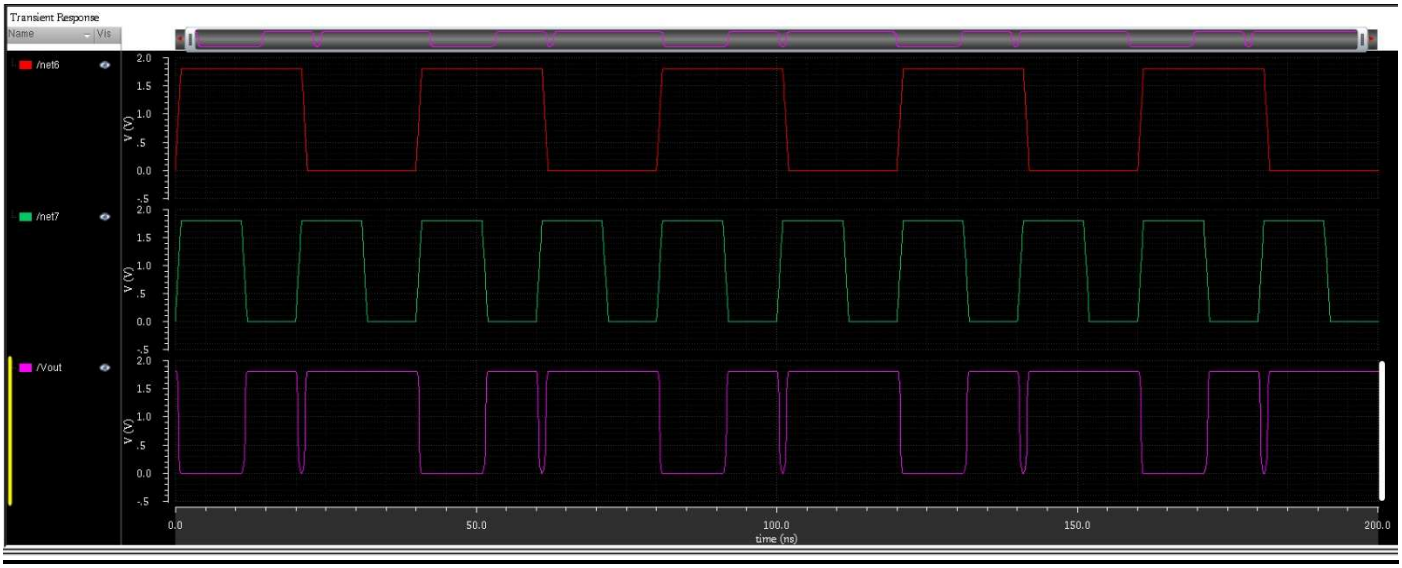


FIG 8.4 Graph of nand gate

OBSERVATION:

The transient simulation shows that the output is LOW when either or both inputs are HIGH. The output becomes HIGH only when both inputs are LOW. The simulated output verifies the truth table of the 2-input NAND gate.

RESULT:

The CMOS 2-input NAND gate was successfully simulated in Cadence Virtuoso, and its output was verified using the truth table.

EXPERIMENT – 9

AIM: To simulate CMOS Half Adder Circuit and verify the truth tables.

TOOLS REQUIRED: PC, CADENCE TOOLS.

THEORY:

A CMOS Half Adder is a combinational circuit used to add two 1-bit binary inputs A and B. It produces two outputs: SUM and CARRY.

i. SUM = $A \oplus B$
ii. CARRY = $A \cdot B$

PROCEDURE:

- Create the CMOS Half Adder schematic in Cadence Virtuoso.
- Connect the required CMOS logic gates for SUM and CARRY.
- Apply input signals A and B using suitable voltage sources.
- Generate the symbol and create the testbench.
- Apply the input combinations and perform transient simulation.
- Observe SUM and CARRY waveforms and verify them with the truth table.

CIRCUIT DIAGRAM:

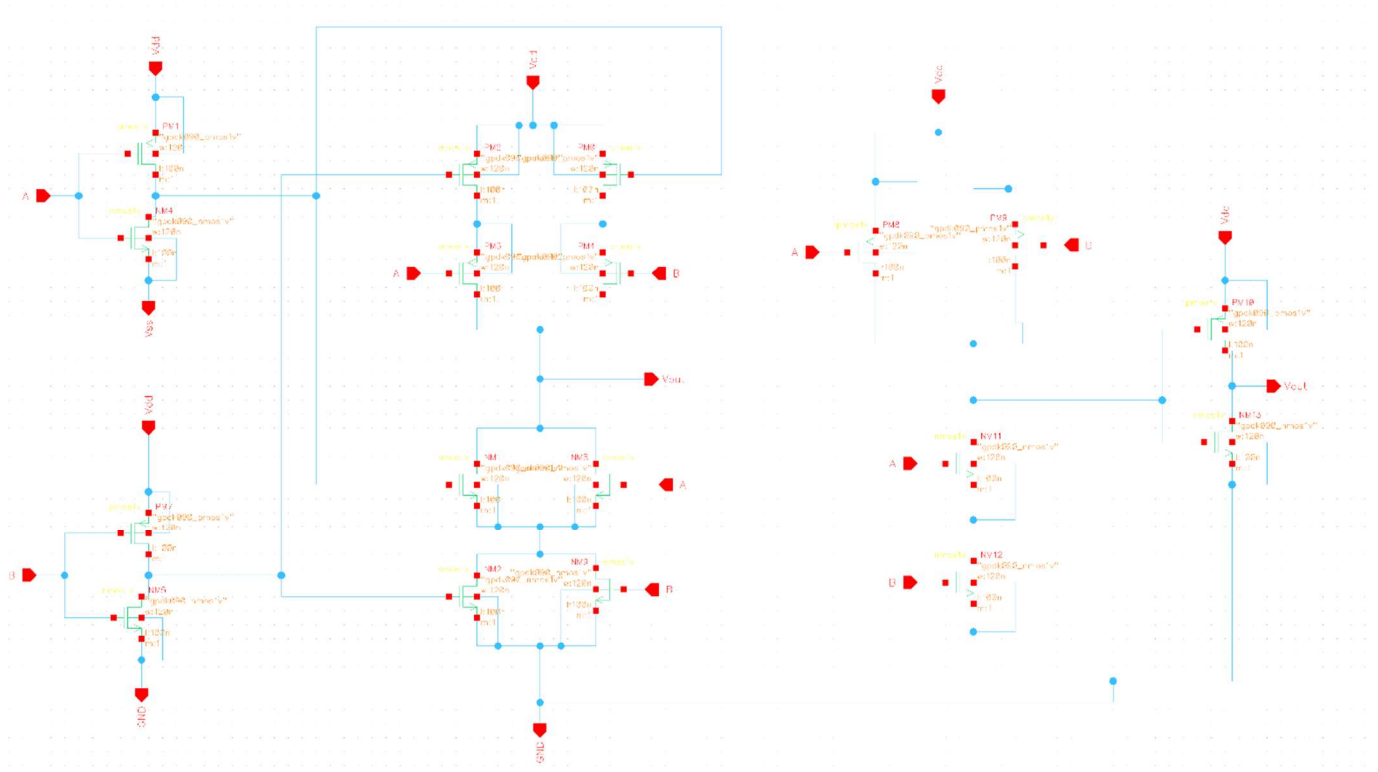


Fig: 9.1 Schematic of Half Adder

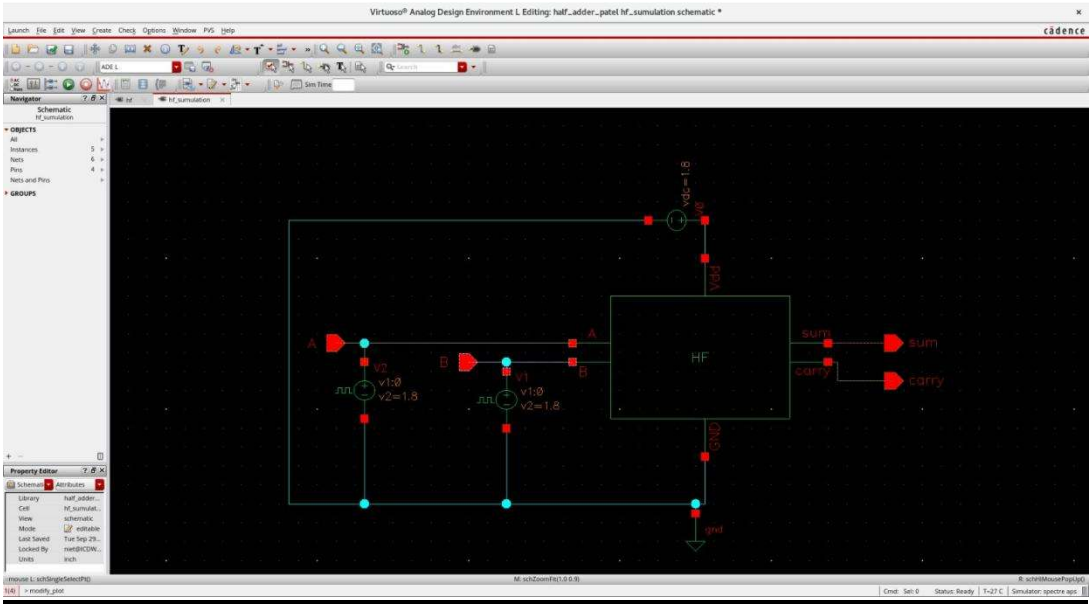


FIG 9.2 pin diagram of Half Adder

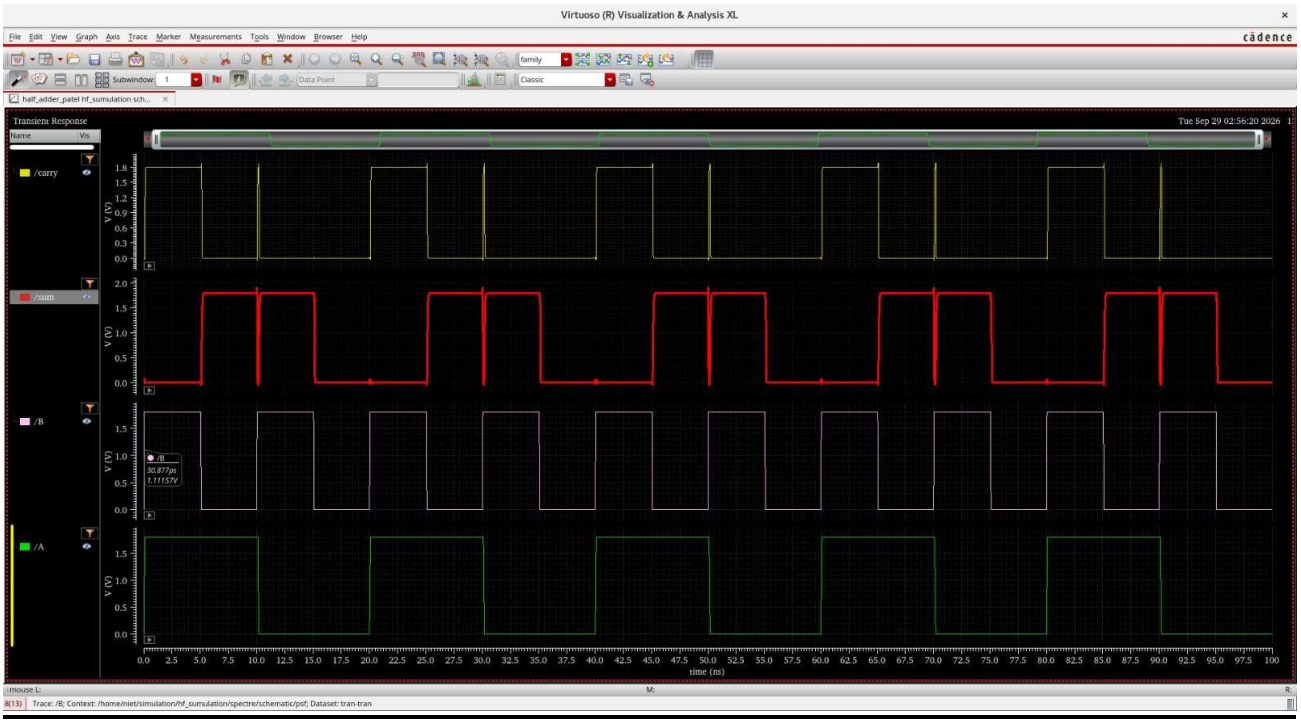


FIG 9.3= Graph of Half Adder

RESULT: The CMOS Half Adder was successfully simulated in Cadence Virtuoso, and the obtained SUM and CARRY outputs verify the expected truth table.

EXPERIMENT - 10

AIM : To simulate the CMOS full Adder Circuit and verify the truth tables

SOFTWARE REQUIRED: Cadence Virtuoso, Spectre simulator, configured CMOS/PDK transistor models, VDD supply and transient (tran) analysis.

THEORY: A full adder is a combinational digital circuit that adds three one-bit binary inputs A, B and C. It produces two outputs: Sum and Carry. The Sum output is the exclusive-OR of the three inputs, while the Carry output represents the carry generated by the addition. The CMOS implementation uses complementary PMOS and NMOS transistor networks to realize the required logic functions.

CIRCUIT DIAGRAM:

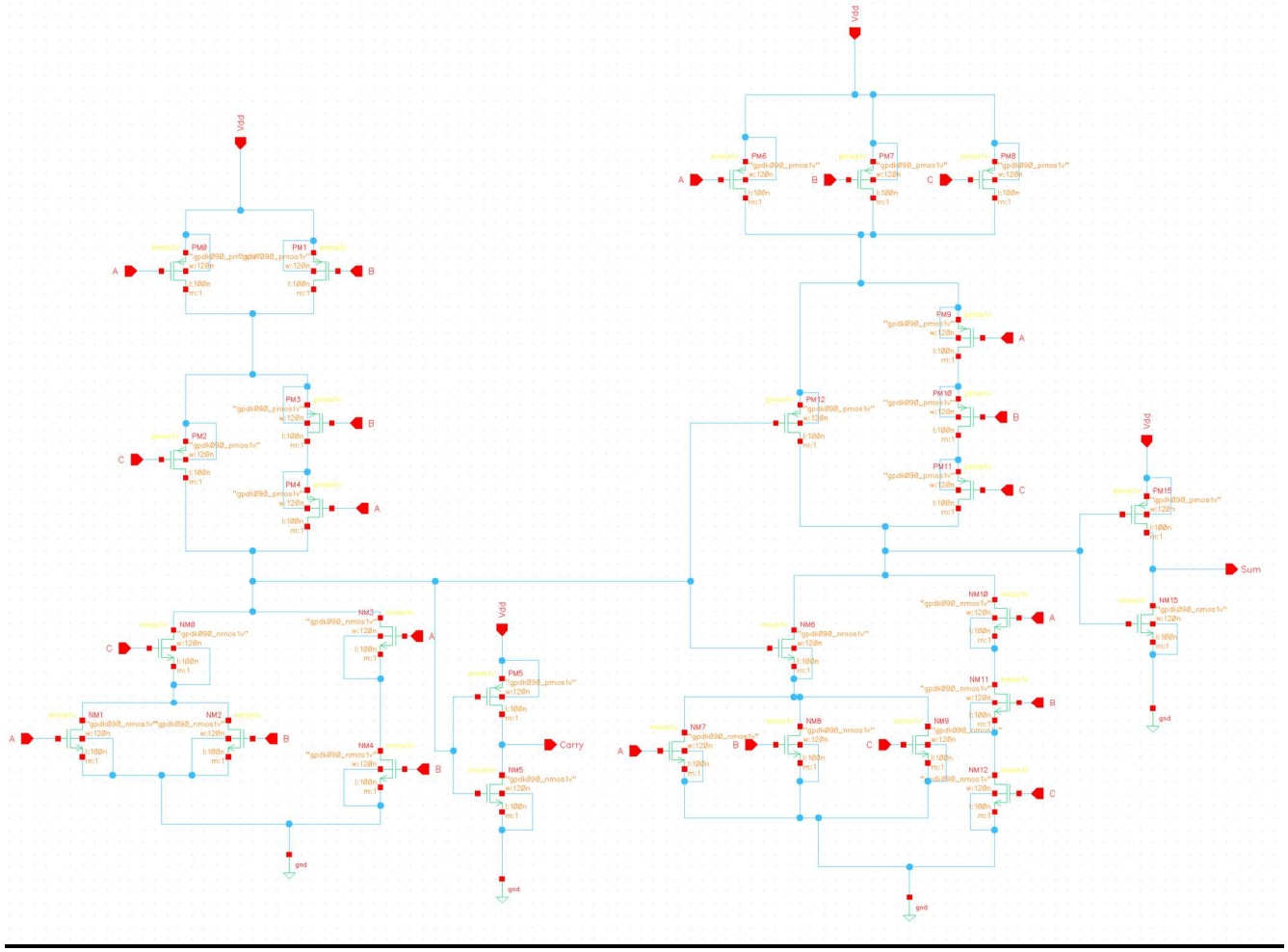


Fig: 10.1 Schematic of Full Adder

Truth Table:

A	B	C	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

PROCEDURE:

1. Create/open the CMOS full adder schematic in Cadence Virtuoso.
2. Connect the three input signals A, B and C to the corresponding transistor networks.
3. Connect VDD and ground and identify the Sum and Carry output nodes.
4. Check the schematic for connectivity errors and save the design.
5. Configure transient analysis in ADE and apply pulse inputs so that all eight input combinations are produced.
6. Plot A, B, C, Sum and Carry and compare the simulated outputs with the truth table.

WAVEFORM:

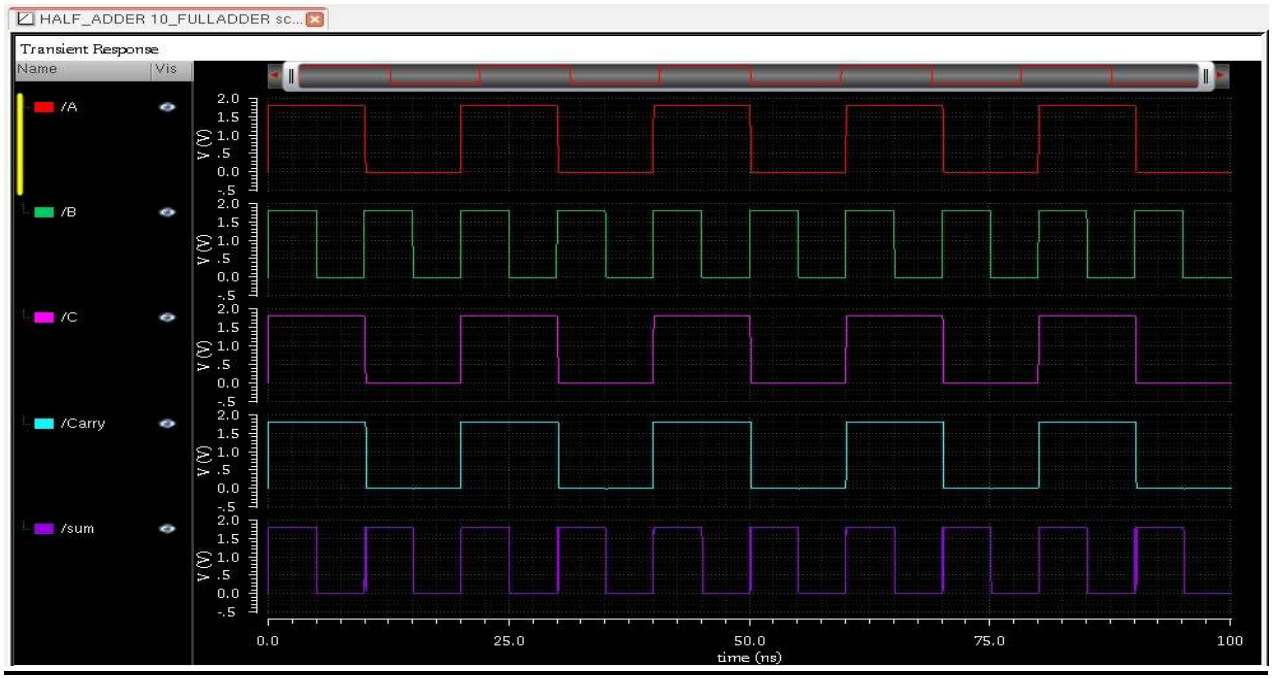


FIG 10.2= Graph of Full Adder

OBSERVATION:

The transient simulation shows the expected Sum and Carry responses for the applied

combinations of A, B and C. The Sum output follows the odd-parity condition, while Carry becomes HIGH for input combinations producing a carry.

RESULT:

The CMOS full adder was successfully designed and simulated in Cadence Virtuoso. The simulated outputs were verified against the full adder truth table.

EXPERIMENT - 11

AIM: To simulate a PMOS transistor switching circuit in Cadence Virtuoso and observe its transient response.

SOFTWARE REQUIRED: Cadence Virtuoso, Spectre simulator, configured CMOS/PDK transistor models, pulse voltage source and transient (tran) analysis.

THEORY: A PMOS transistor conducts when its gate-to-source voltage is sufficiently negative. In the given test circuit, a pulse signal is applied to the gate and the output response is observed during transient analysis. The waveform is used to examine the switching behavior of the circuit.

CIRCUIT DIAGRAM:

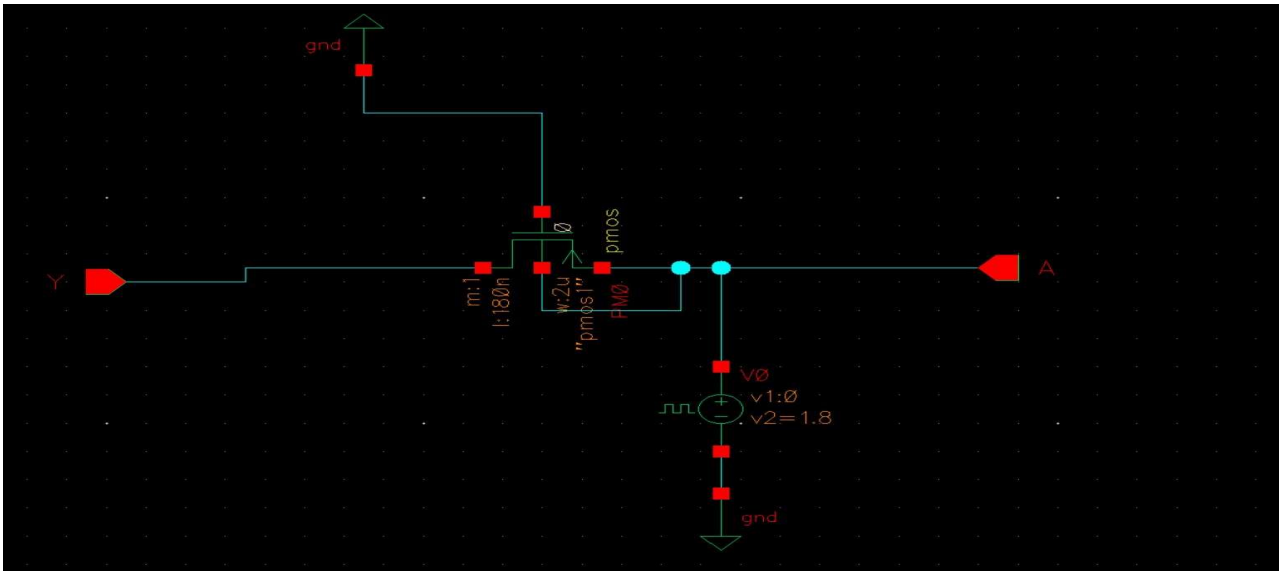


Figure 11.1: PMOS transistor test schematic in Cadence Virtuoso

PROCEDURE:

1. Create a new schematic cell view in Cadence Virtuoso.
2. Place the PMOS transistor and pulse voltage source, then connect the circuit as shown.
3. Connect the required ground reference and label the input and output nodes.
4. Check and save the schematic, then open ADE for simulation.
5. Select transient analysis, run the simulation and plot the input and output waveforms

TRANSIENT RESPONSE:

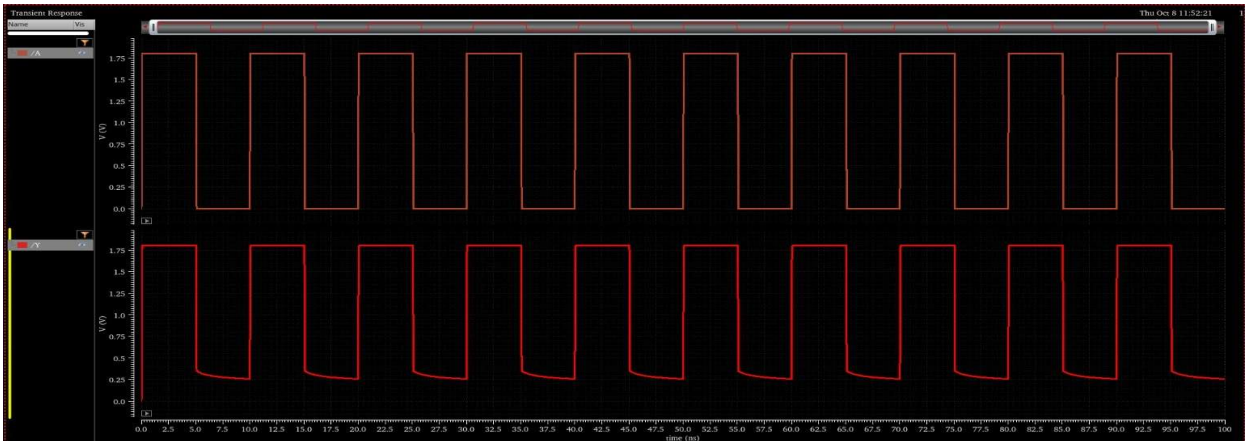


Figure 11.2: Transient response of the PMOS test circuit

OBSERVATION: The input signal switches between approximately 0 V and 1.8 V. The output follows the switching pattern, with a non-ideal low-level response visible in the plotted waveform.

RESULT: The PMOS transistor test circuit was simulated in Cadence Virtuoso and its transient switching response was observed.

EXPERIMENT - 12

AIM: To design and simulate a CMOS Transmission Gate (TG) in Cadence Virtuoso and verify its operation using transient analysis.

SOFTWARE REQUIRED: Cadence Virtuoso, Spectre simulator, configured CMOS/PDK transistor models, pulse and DC voltage sources, and transient (tran) analysis.

THEORY: A CMOS transmission gate is a bidirectional switch formed by connecting one NMOS and one PMOS transistor in parallel. Their gate-control signals are complementary. When the transmission gate is enabled, the input signal is passed to the output; when disabled, the path is turned off. The parallel NMOS and PMOS devices help pass both logic-low and logic-high levels effectively.

CIRCUIT DIAGRAM:

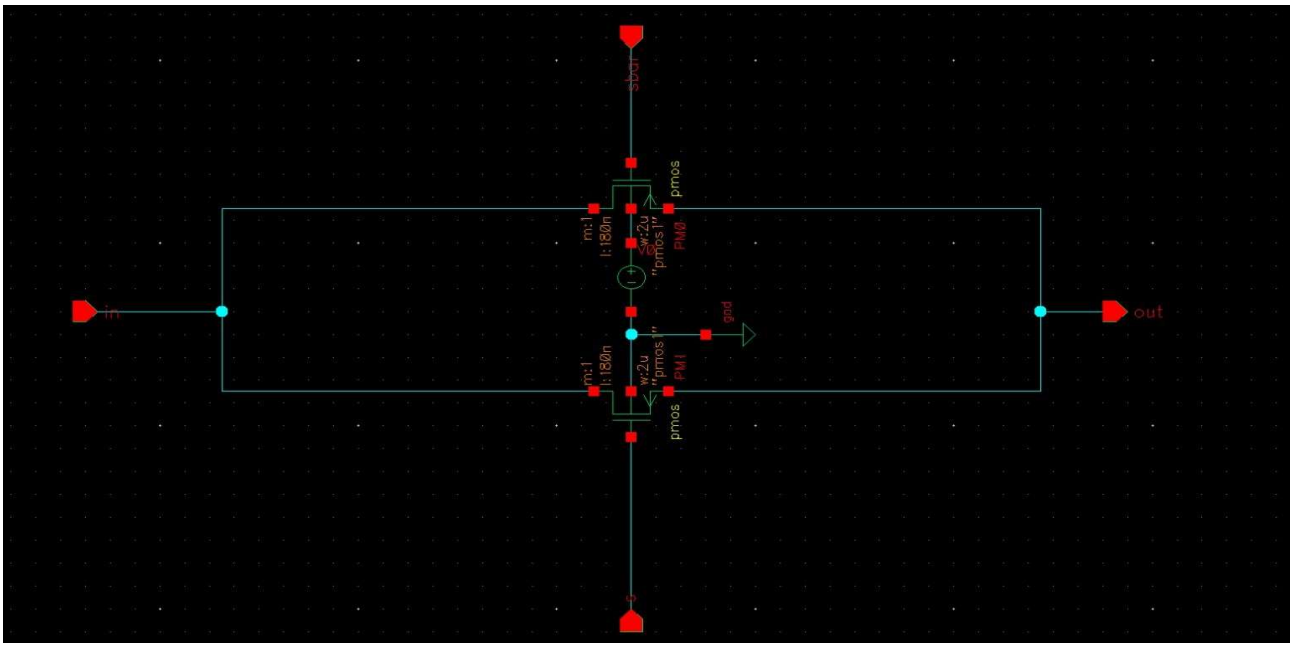


Figure 12.1: CMOS transmission gate transistor-level schematic

PROCEDURE:

1. Create a new schematic cell view for the transmission gate.
2. Place one NMOS and one PMOS transistor and connect their source-drain paths in parallel.
3. Apply complementary control signals to the transistor gates and connect the input and output terminals.
4. Add the required voltage sources and ground references, then check and save the schematic.

TRANSMISSION GATE SYMBOL AND TEST CIRCUIT:

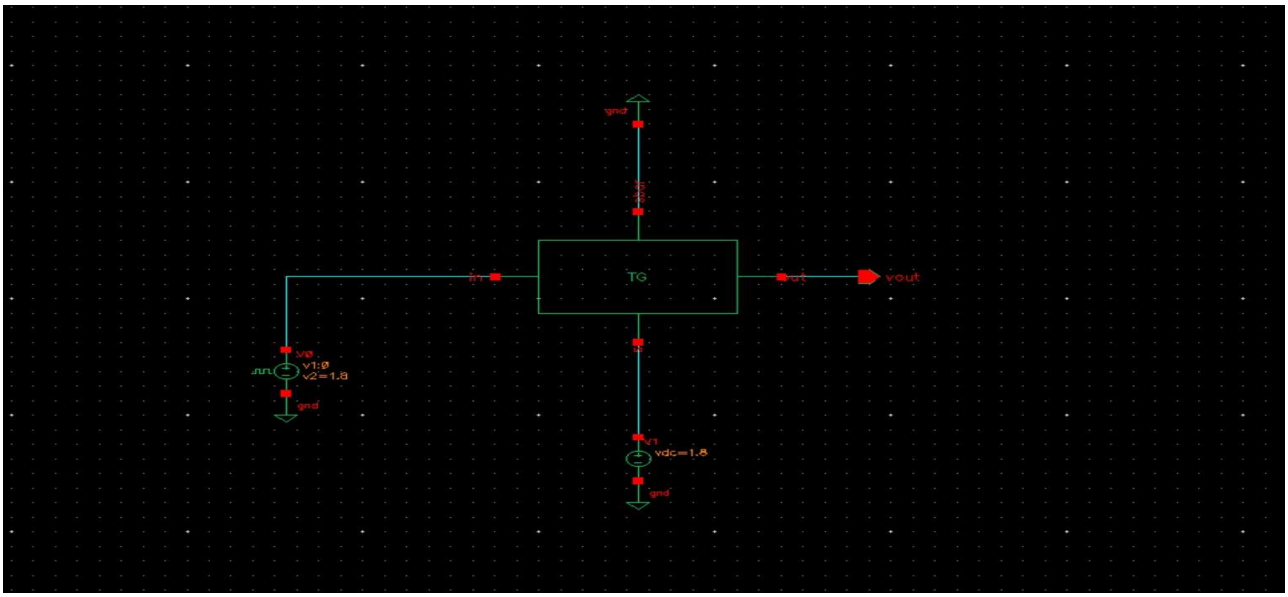


Figure 12.2: Transmission gate block symbol

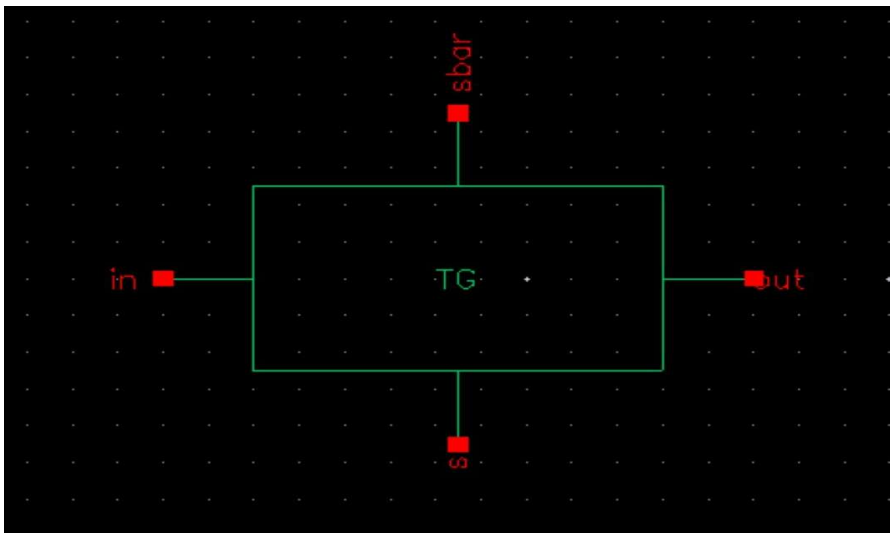


Figure 12.3: Transmission gate test circuit in Cadence Virtuoso

SIMULATION: Open ADE, select transient analysis and set the stop time to cover the complete input pulse sequence. Select the input and output nets for plotting, run the simulation and compare the waveforms.

WAVEFORM:

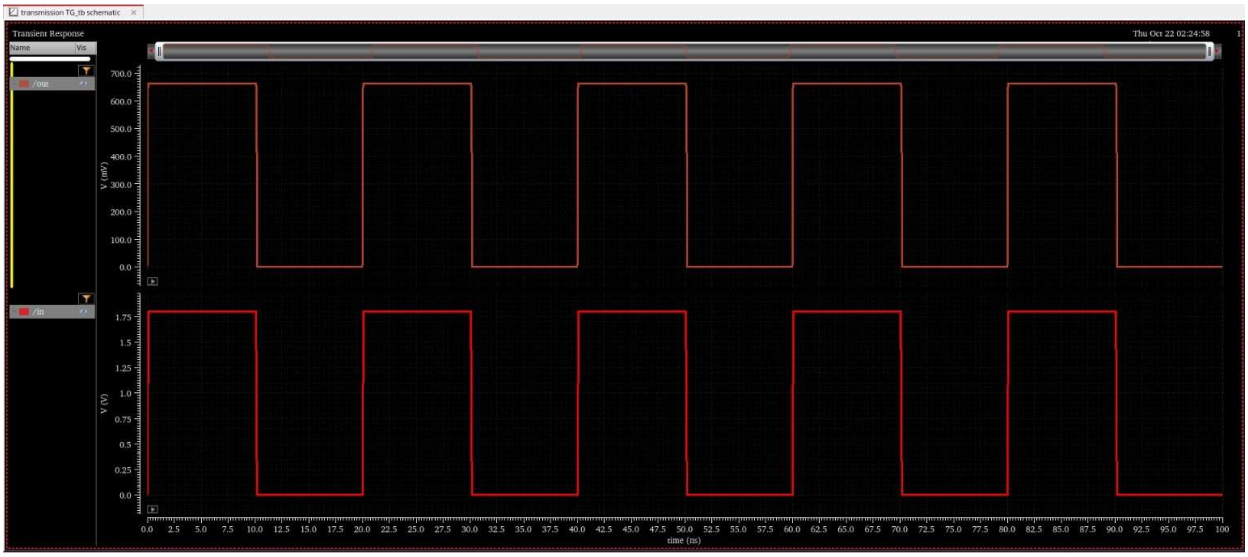


Figure 12.4: Transient response of the CMOS transmission gate

OBSERVATION: The plotted input switches between approximately 0 V and 1.8 V, and the output follows the applied input pattern during the conducting interval. The waveform demonstrates signal transfer through the transmission gate under the applied control condition.

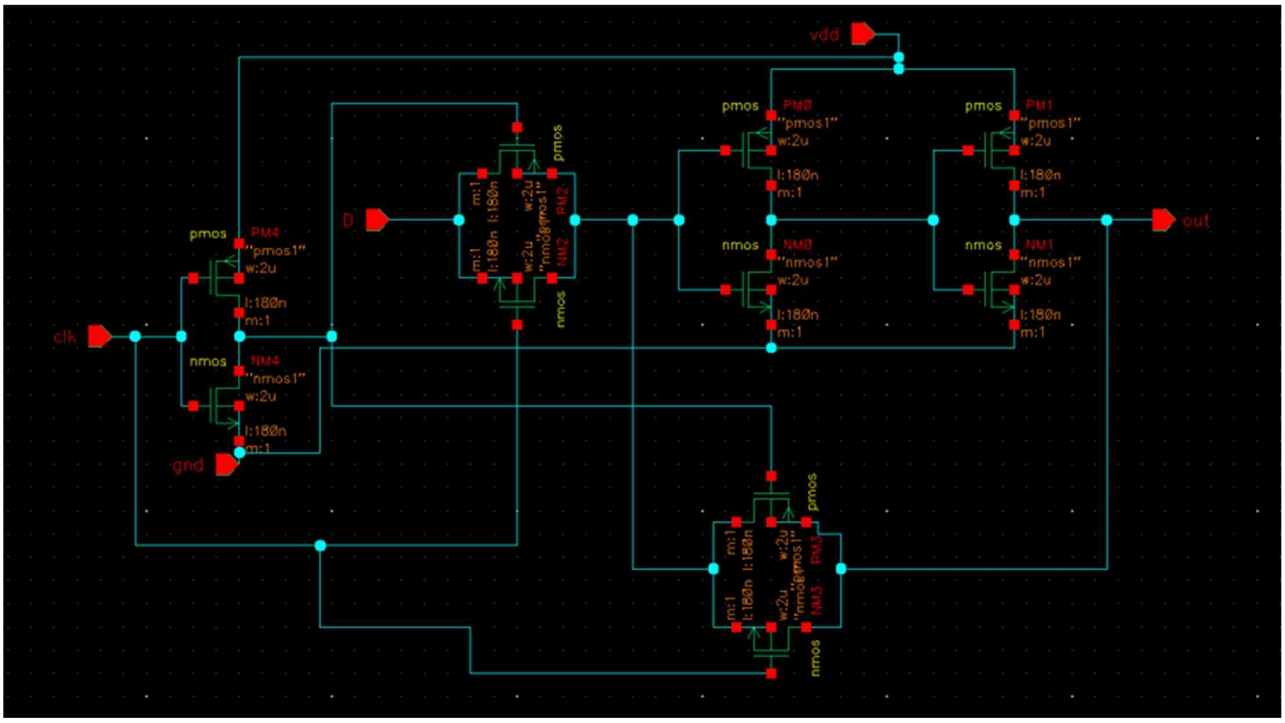
RESULT: The CMOS transmission gate schematic was constructed and its transient response was observed in Cadence Virtuoso

EXPERIMENT – 13

AIM: To design and simulate a CMOS D-Flip Flop in Cadence Virtuoso and verify its operation using transient analysis.

THEORY: A D-Flip Flop is a sequential logic circuit that stores one bit of data. It transfers the input data (D) to the output (Q) on the active edge of the clock signal. The output remains unchanged between active clock edges. The complementary output is represented by $\bar{Q}$.

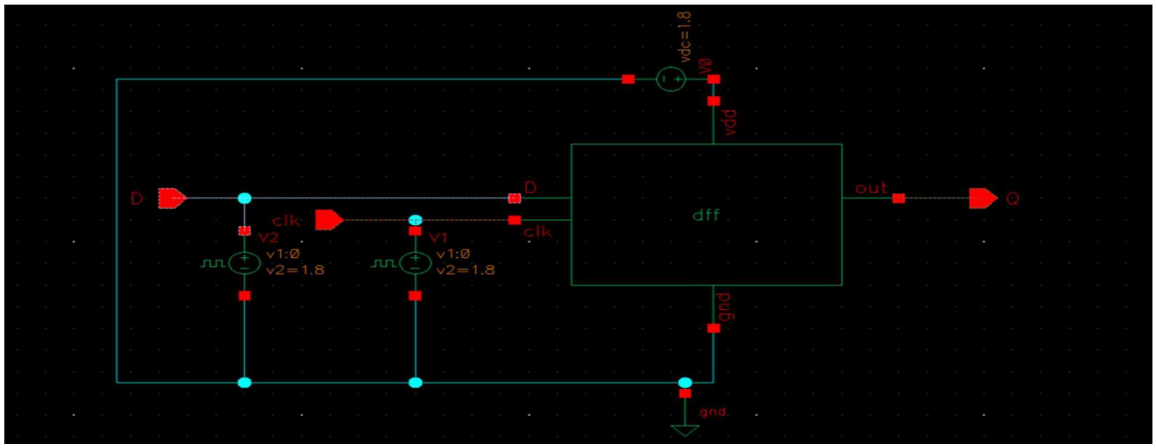
CIRCUIT DIAGRAM:



PROCEDURE:

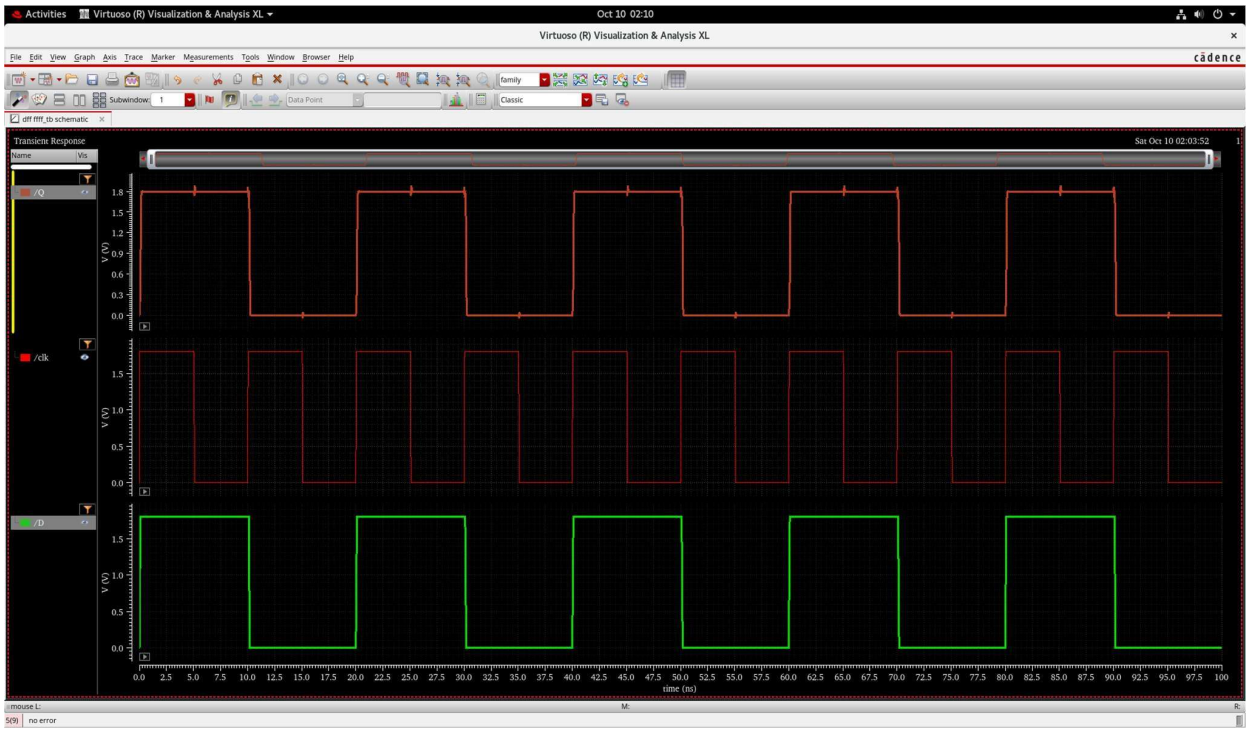
1. Create a new schematic cell view for the D-Flip Flop.
2. Place the required CMOS transistors and connect them according to the D-Flip Flop circuit.
3. Connect the input data (D), clock (CLK), output (Q), and complementary output ($\bar{Q}$), if required.
4. Add the required voltage sources and ground references, then check and save the schematic.

D-FLIP FLOP SYMBOL AND TEST CIRCUIT:



SIMULATION: Open ADE, select transient analysis, and set the stop time to cover the complete input and clock pulse sequence. Select the input, clock, and output nets for plotting, run the simulation, and compare the waveforms.

WAVEFORM:



OBSERVATION: The output Q follows the input D at the active edge of the clock signal and holds its previous between active clock edges. The waveform verifies the data storage and transfer operation of the D-Flip Flop.

RESULT: The CMOS D-Flip Flop schematic was constructed and its transient response was observed in Cadence Virtuoso.

EXPERIMENT – 14

Aim: To design and simulate a CMOS inverter and verify the layout using DRC and LVS checks**.**

Software Required: Cadence Virtuoso & GPDK 90 nm Technology Library.

Theory: A CMOS inverter is a basic digital circuit consisting of one PMOS and one NMOS transistor connected in series.

- The PMOS transistor is connected between VDD and the output.
- The NMOS transistor is connected between the output and GND.
- Both transistor gates are connected together to form the input.
- Their drains are connected together to form the output.

When the input is LOW, the PMOS turns ON and the NMOS turns OFF, producing a HIGH output.
When the input is HIGH, the PMOS turns OFF and the NMOS turns ON, producing a LOW output.
Thus, the output is the logical complement of the input**.**

Circuit Design and Schematic:

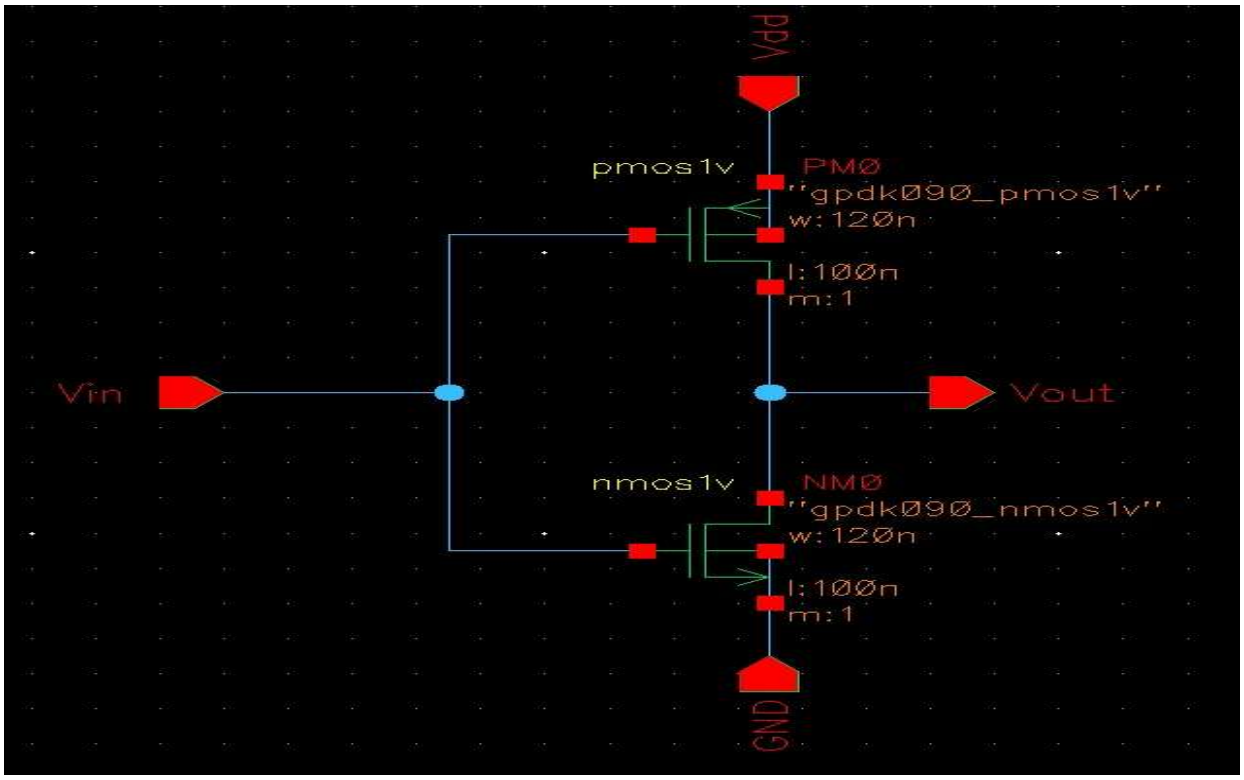


Figure 1: CMOS Inverter Schematic

The CMOS inverter schematic was designed using PMOS and NMOS transistors from the GPDK 90 nm library.

Symbol Creation:

After completing the schematic, a symbol view of the CMOS inverter was created.

The symbol contains:

- Input terminal: Vin
- Output terminal: Vout

- Power terminal: VDD
- Ground terminal: GND

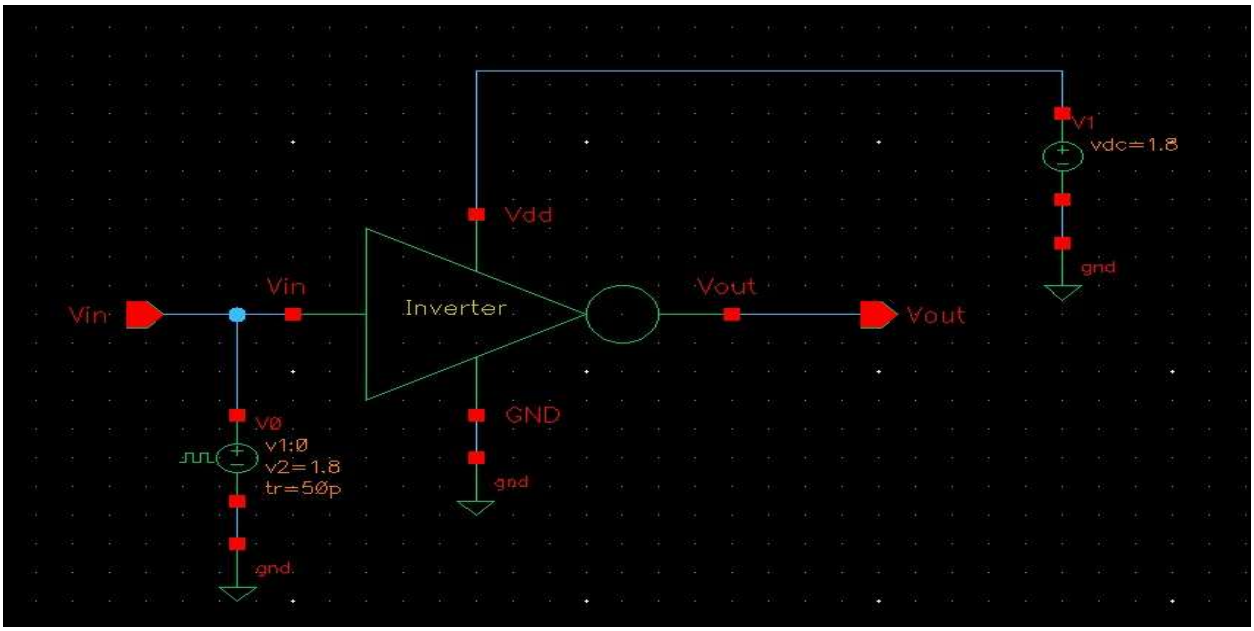


Figure 2:CMOS Inverter Symbol

DC and Transient Analysis

Graph: DC Analysis

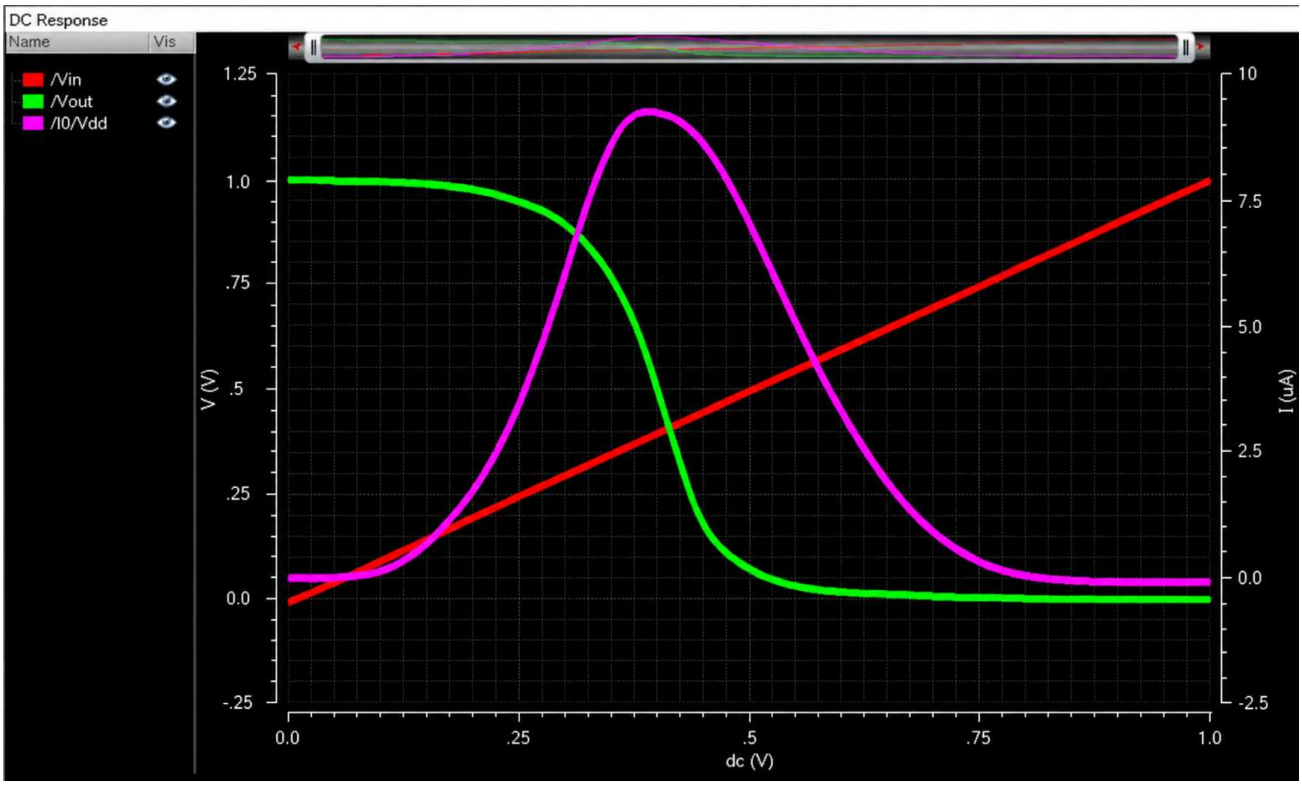


Figure 3: DC Analysis Graph

Waveform:

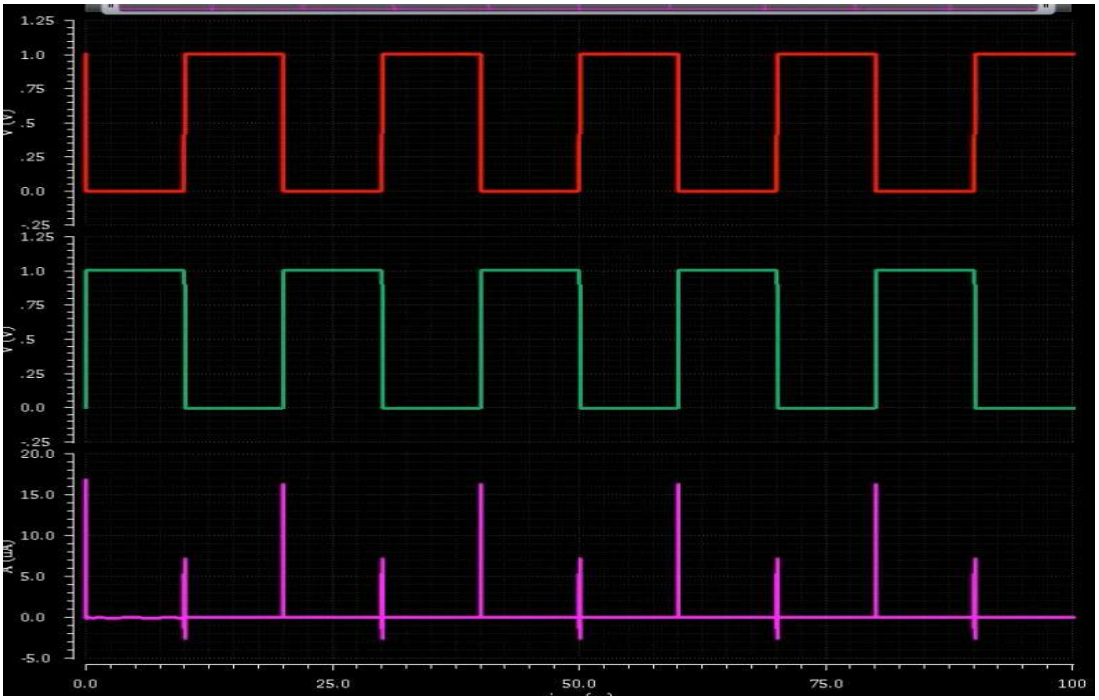


Figure 4:Tran. Analysis Waveform

CMOS Inverter Layout

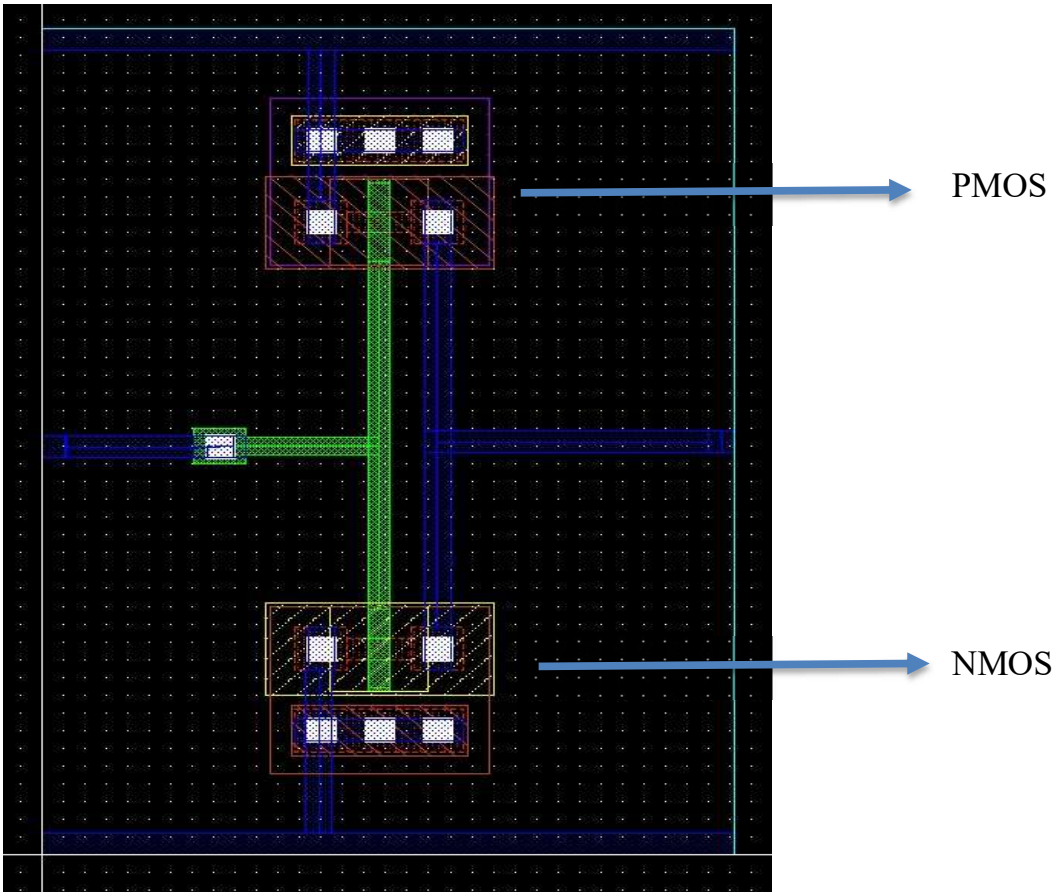


Figure 5: CMOS Inverter Layout

Design Rule Check (DRC):

DRC was performed to verify whether the physical layout follows the fabrication design rules of the selected technology.

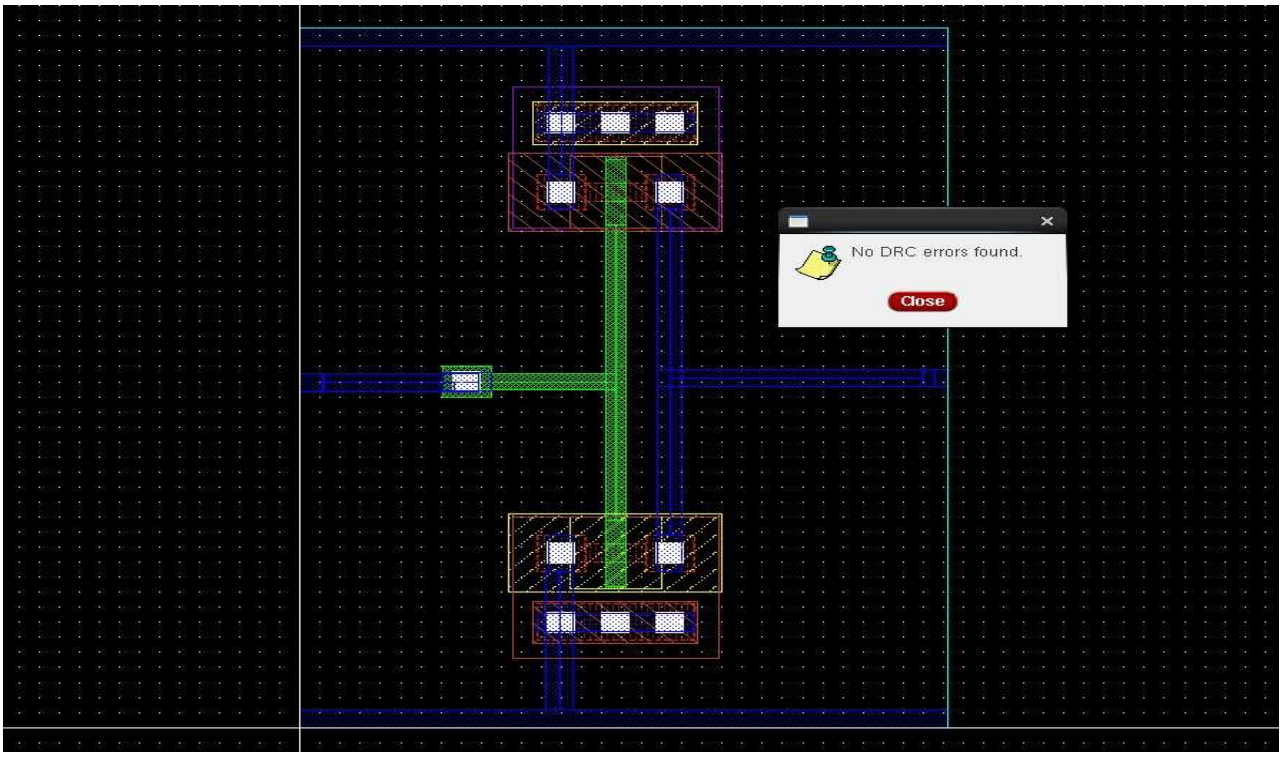


Figure 6: NO DRC Found Window (DRC Successful)

The DRC verification window displayed: **"No DRC errors found."**

This confirms that the layout satisfies the checked design rules**.**

Layout Versus Schematic (LVS):

LVS verification was performed to compare the physical layout with the original schematic.

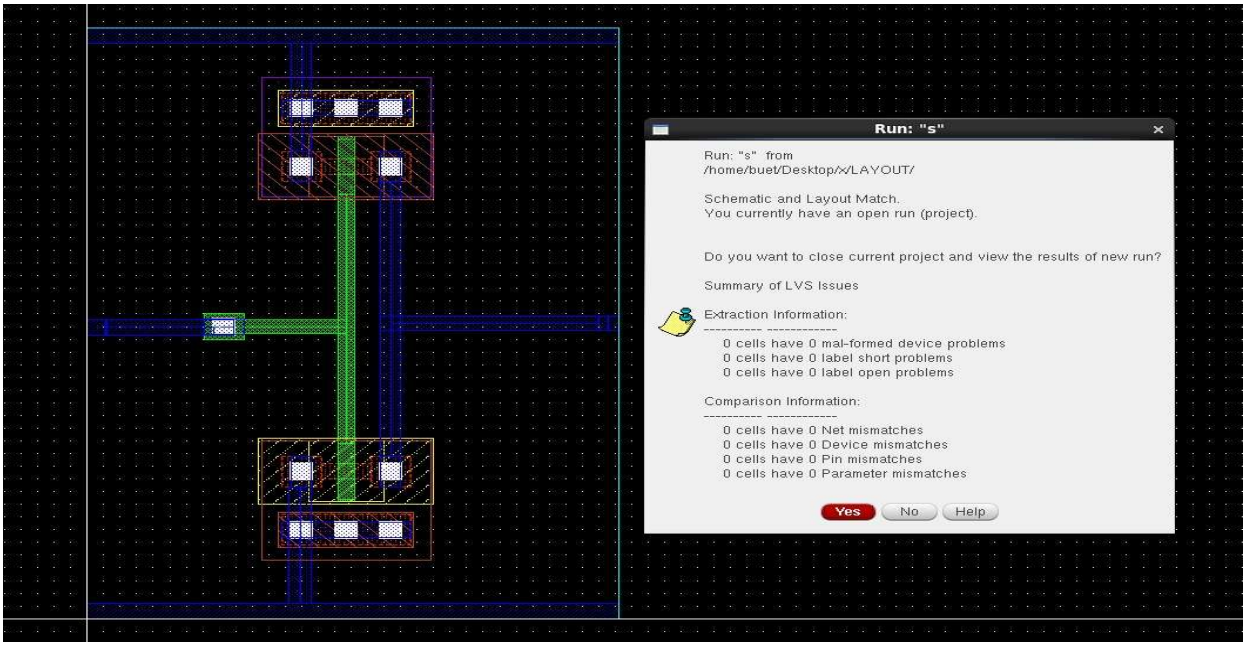


Figure 7: zero errors found window

Observation: The LVS Debug window displayed.

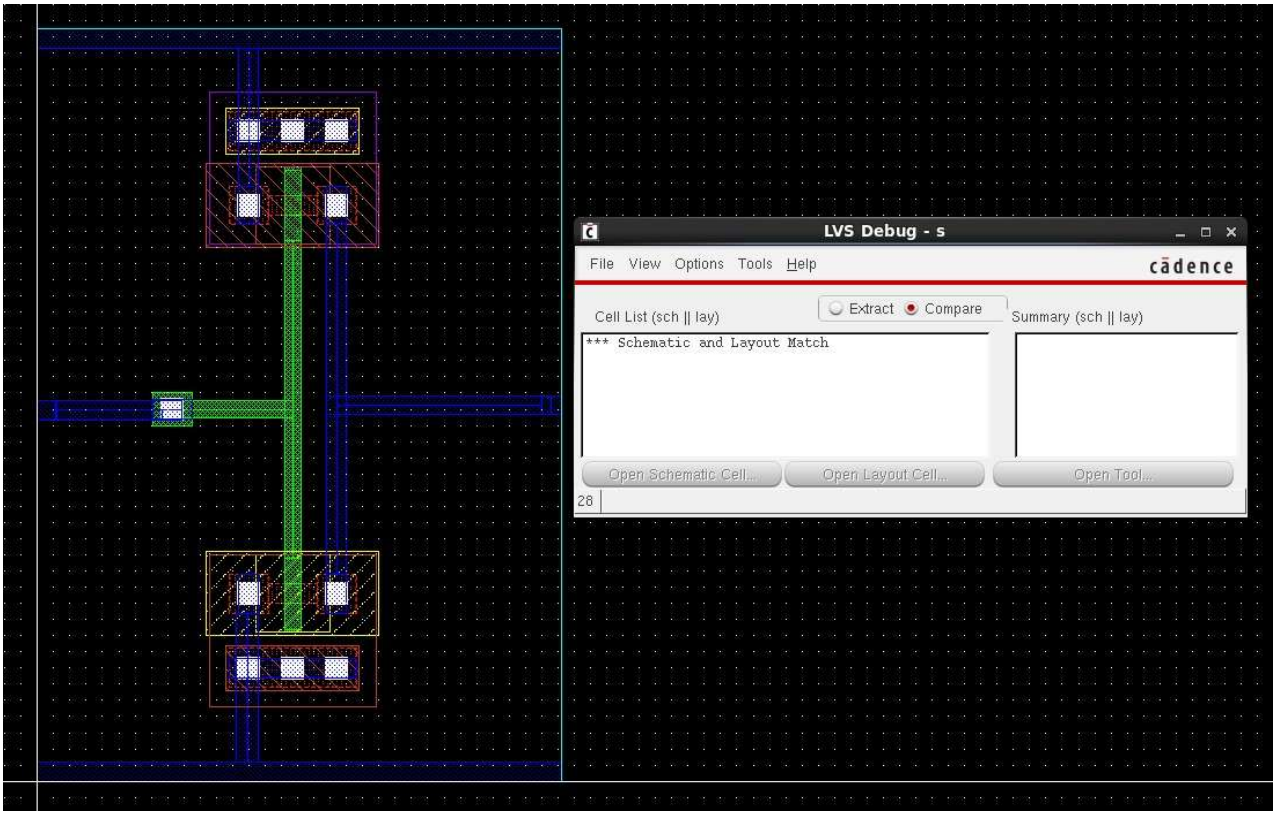


Figure 8: Schematic & Layout Matched

Schematic and Layout Matched.

CONCLUSION:

The CMOS inverter was successfully designed using PMOS and NMOS transistors in Cadence Virtuoso. Its symbol was created, and DC and transient analyses verified the expected inverting operation. The physical layout was also completed successfully. DRC reported no errors, and LVS confirmed that the schematic and layout match.

RESULT:

The CMOS inverter was successfully designed, simulated, and laid out using Cadence Virtuoso. The expected inverter characteristics were verified through DC and transient analysis. The layout passed DRC verification with no errors and LVS verification with a successful schematic and layout match.

EXPERIMENT – 15

AIM: To Design and Simulate a 6 – Transistor SRAM cell.

Software Used

- Cadence Virtuoso 6.1.5
- GPDK 90 nm Technology Library

Theory

A 6T SRAM cell is a Static Random-Access Memory cell made up of six MOS transistors. It consists of two PMOS transistors and four NMOS transistors.

Four transistors form two cross-coupled CMOS inverters that store one bit of data. The remaining two NMOS transistors act as access transistors and connect the internal storage nodes to the bit lines.

The cell uses the following signals:

- WL (Word Line): Controls access to the cell.
- BL (Bit Line): Transfers data.
- BLB (Bit Line Bar): Complementary bit line.
- Q and QB: Complementary storage nodes.

The cross-coupled inverters retain the stored data as long as power is supplied.

Schematic Design

The 6T SRAM cell schematic was designed in Cadence Virtuoso using the GPDK 90 nm technology library.

The two internal storage nodes are labelled Q and QB.

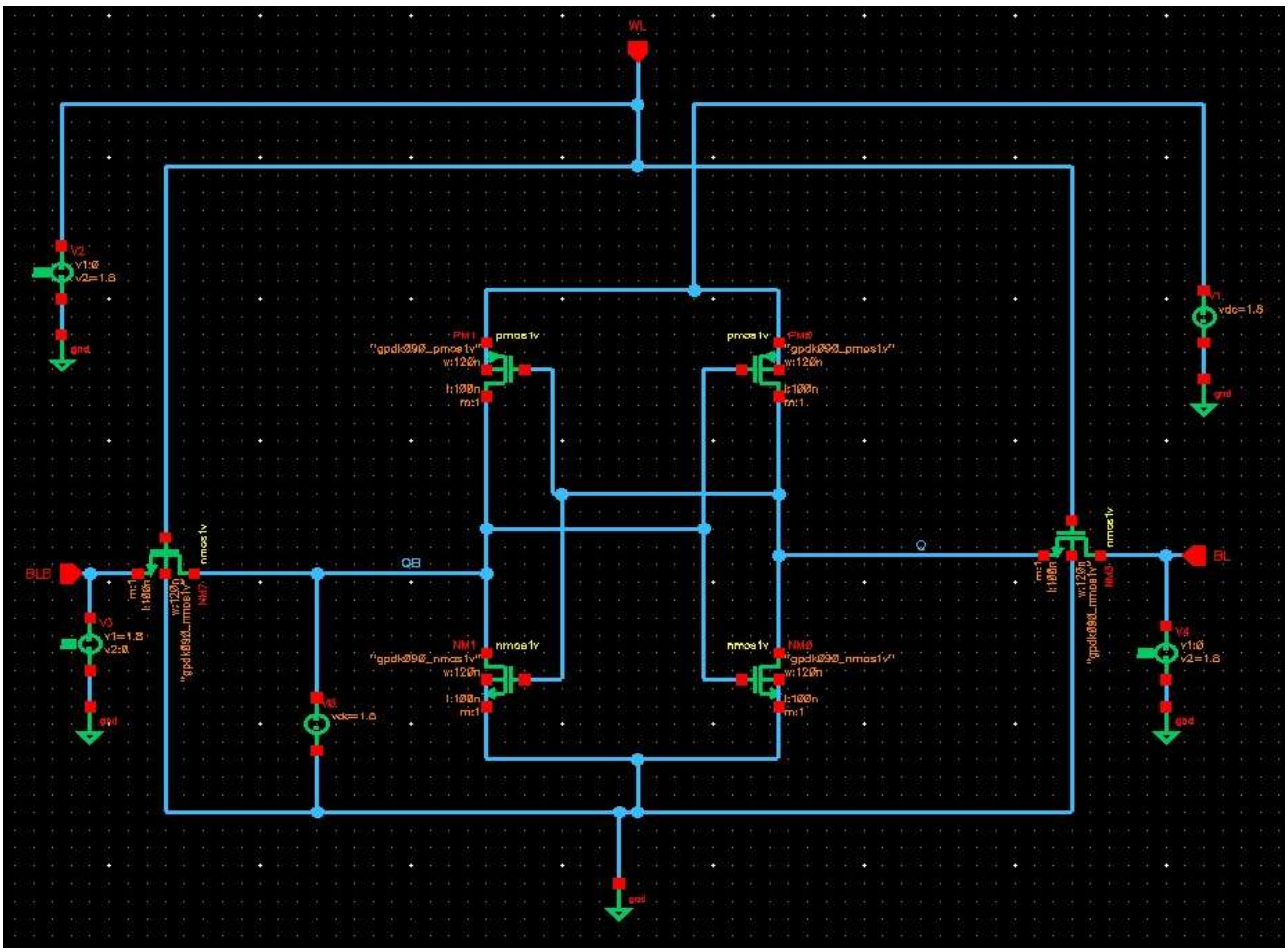


Figure 1:Schematic Diagram of 6T SRAM Cell Designed in Cadence Virtuoso.

Simulation Setup

The schematic was simulated using the Spectre simulator in Cadence Virtuoso.

- Analysis Type: DC Sweep
- Sweep Variable: V0
- Start Voltage: 0 V
- Stop Voltage: 1.8 V
- Supply Voltage: 1.8 V
- Output Signals: Q and QB

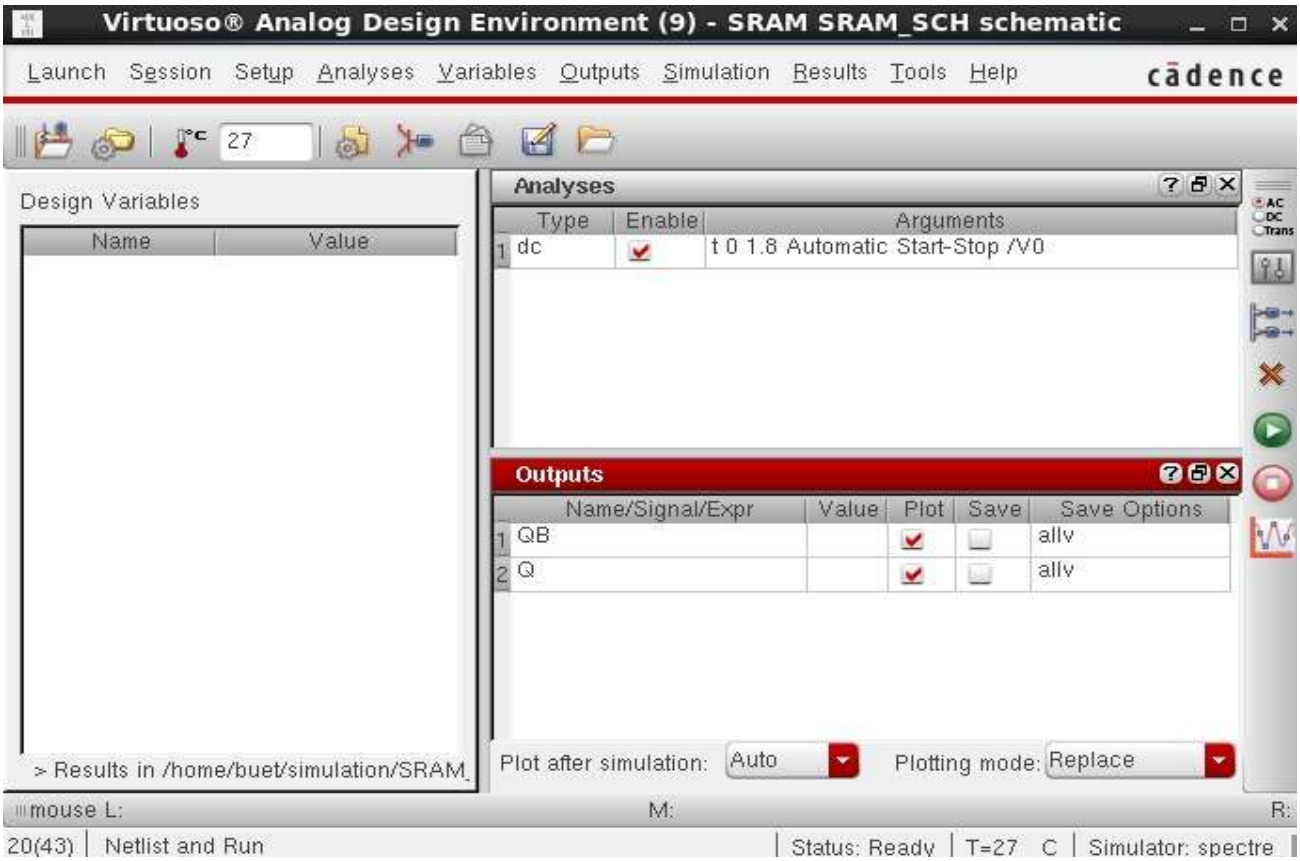


Figure 2: DC Analysis Setup for 6T SRAM Cell

DC Analysis: The voltage variations at the internal storage nodes Q and QB were observed.

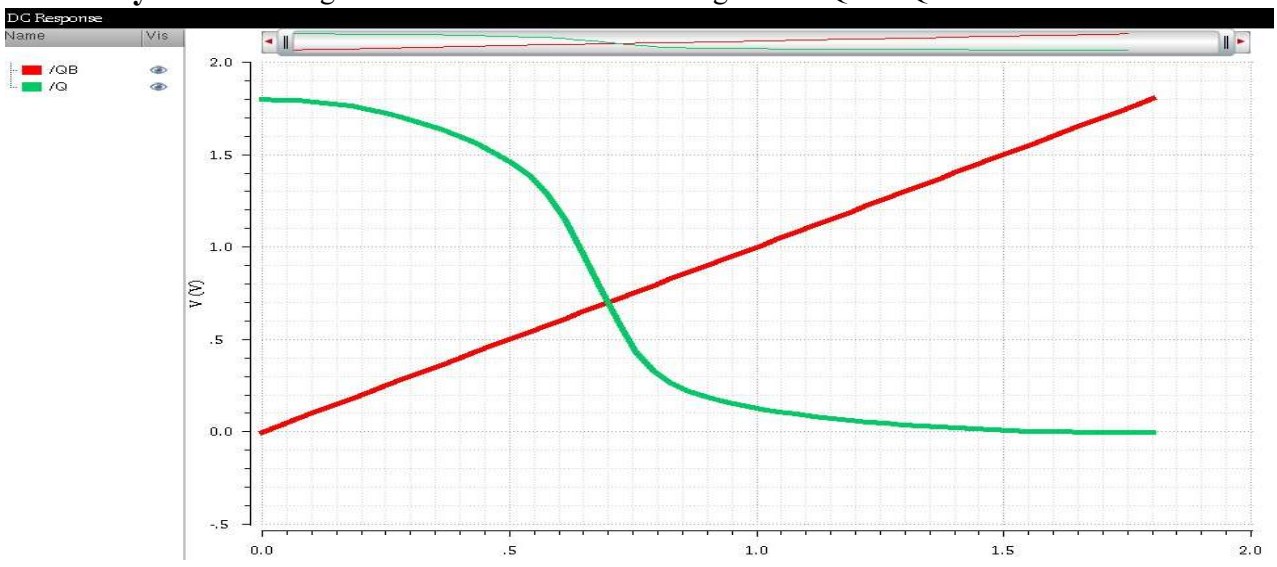


Figure3: DC Response Showing the Complementary Voltage Characteristics of Storage Nodes Q and QB.

SRAM Butterfly Curve

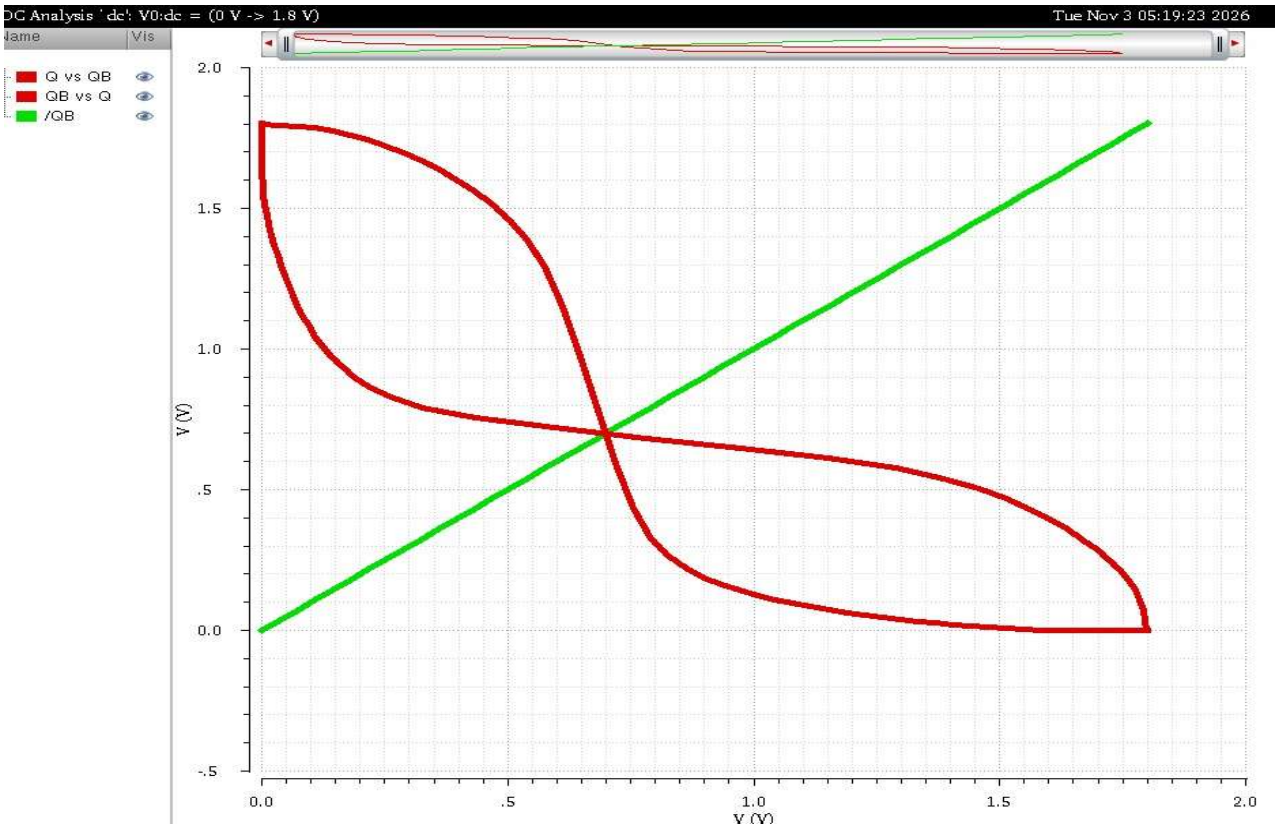


Figure4: DC Response Showing the Voltage Characteristics of Nodes Q and QB.

RESULT: The 6T SRAM cell was successfully designed and simulated in Cadence Virtuoso. The DC response and butterfly curve were obtained successfully.

CONCLUSION: The DC characteristics of the 6T SRAM cell were analysed. The complementary behaviour of storage nodes Q and QB was observed. The butterfly curve was obtained to study the stability characteristics of the SRAM cell.